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Title: **Does Coverage Expansion Grow the Physician Pipeline? Evidence from the ACA**

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ABSTRACT

Does insurance expansion grow the physician training pipeline? We study how the ACA's Medicaid expansion changed residency training using a twenty-year institution panel of the National Resident Matching Program (2000–2019) that we construct. Number of residency spots at existing institutions falls by an imprecise 6 percent, and offered positions barely move.

JEL: I11, I13, I18, J44, H75

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MANUSCRIPT TEXT

1 Introduction

The geographic distribution of physicians shapes the population's health in measurable ways. A substantial body of evidence links primary care supply to improvements in life expectancy, disease-specific mortality, and preventable hospitalizations. Basu et al. (2019) estimate that each ten-physician increase in primary care supply per 100,000 population is associated with a 51.5-day gain in life expectancy and meaningful reductions in cardiovascular and cancer mortality. Macinko, Starfield, and Shi (2007) reach similar conclusions using state-level variation in primary care density, finding lower mortality rates for heart disease, cancer, and stroke in states with more primary care physicians. The relationship is not confined to the United States: evidence from Germany (Sundmacher and Busse 2011) and Canada (Piérard 2014) likewise documents health gains from improved access to general practitioners, suggesting the pattern reflects a general feature of primary care rather than an artifact of the US institutional context. Because where physicians complete residency training strongly predicts where they ultimately practice (LeFevre and Colwill 1983; Murphy 2020; Orr 2023), constraints on residency capacity translate into downstream constraints on the geographic distribution of physician supply. In this paper, we ask whether state-level Medicaid expansion changed the number of residency positions that hospitals offer and fill.

Whether policies designed to expand physician supply reliably produce these health benefits depends heavily on institutional context and program design. Evidence from earlier Medicaid expansions establishes that insurance coverage expansions can shift physician supply toward underserved areas through financial incentives. Qian (2024) show that 1980s expansions for pregnant women increased the number of OBGYNs in low-income counties, and Huh and Lin (2024) find that OBGYN counts per capita rose and remained persistently elevated following these expansions, with early-career physicians accounting for much of the growth. Parallel effects appear in dental care: Huh (2021) find that dental Medicaid expansions raised dentists per capita in poorer counties by 13 percent, and Buchmueller, Miller, and Vujicic (2016) document corresponding increases in dentist participation and patient access. Baker and Royalty (2000) further show that Medicaid-driven demand shocks were absorbed primarily by public hospitals and clinics rather than private practices, pointing to institutions as the key margin of adjustment. This last finding is particularly relevant to residency training, where hospitals both collect the expanded Medicaid revenue and decide how many training slots to fund.

International evidence introduces important qualifications. Carrillo and Feres (2019) evaluate a Brazilian initiative to increase primary care physician supply and find that while the program raised the volume of physician visits, it did not improve infant health outcomes, a pattern consistent with physicians substituting for nurses rather than generating new clinical value. A related evaluation of Brazil's More Doctors Program finds increases in healthcare utilization and reductions in preventable hospitalizations, though all-cause mortality remained unchanged (Mattos and Mazetto 2019). In Japan, Iizuka and Watanabe (2016) show that a national residency reform intended to rationalize training pathways had the unintended consequence of drawing physicians away from rural areas, contributing to hospital closures as institutions faced higher wage pressures to attract and retain residents. Taken together, these experiences indicate that expanding physician training capacity is a necessary but not sufficient condition for improving health: the translation from training slots to health outcomes depends on where newly trained physicians choose to practice, the complementary inputs available in those settings, and the financial incentives governing provider behavior.

Did ACA Medicaid expansion change the number of residency positions hospitals offer and fill? We answer it using the residency programs that fill their positions through the national match. The National Resident Matching Program (NRMP) records the number of positions every program offers and the number it fills, which we link to the staggered timing of state Medicaid expansions under the Affordable Care Act. Our panel spans two decades, 2000 to 2019. Because states expanded Medicaid in different years beginning in 2014, we estimate the effect with a staggered difference-in-differences event study, using the imputation estimator of Borusyak, Jaravel, and Spiess (2024).

Our central finding is that training supply did not respond to expansion. At existing training institutions—the intake margin—expansion is followed by an average decline of 0.014 matched positions per 100,000 population, about 5.7 percent of the pre-expansion mean. This estimate is statistically indistinguishable from zero. Offered positions barely move (−3 percent, $p = 0.21$), so the margin at issue is slot creation rather than unfilled slots. A flat intake margin could still conceal a supply response if expansion induced hospitals that had never trained residents to open programs, and the raw match data appear to show exactly that: 201 sponsoring institutions enter the NRMP after 2010, disproportionately in expansion states. Most of that entry, however, is an artifact of accreditation reform rather than new training capacity. Our classification shows that roughly half of the post-2010 entrants are pre-existing American Osteopathic Association (AOA)-accredited sponsors—institutions that were already training residents under osteopathic accreditation and that appear in the match only because the single-accreditation transition moved them into the NRMP after 2015. Among expansion-state entrants in the 2016–2019 window, the share is 63 percent, because osteopathic training was historically concentrated in states that expanded Medicaid. Once entrants are classified, genuinely new sponsors show no differential entry response to expansion. State-level training totals, which include the entry margin, do not decline after expansion, with or without the migrating institutions. A placebo that shifts every state's expansion back ten years, estimated on pre-ACA data alone, comes back null. Because treatment is assigned at the state level, few-cluster randomization inference, wild cluster bootstrap, and the Rambachan and Roth (2023) sensitivity analysis leave every individual estimate imprecise, and we report those standards uniformly. The consistent pattern across margins—flat intake at existing institutions, no genuine entry response, and non-declining state totals—indicates that the training pipeline did not respond to the largest insurance expansion in half a century.

Our contribution is threefold. First, we provide the first evidence on whether a coverage expansion grows the physician training pipeline. We find that it does not: intake at existing institutions is flat, genuinely new entry does not respond, and state totals do not decline. This finding matters for policy because the insured population and the demand for physicians are both growing, and projections of physician shortages assume that training will expand to meet them. Our results indicate that coverage policy alone does not generate that expansion; the policies that moved training capacity over this period were direct interventions rather than coverage-induced changes in hospital finances. Second, the decomposition reconciles the hospital-level and state-level evidence that previously pointed in opposite directions, including the state-level aggregates in Ang and Beniwal (2026). Third, we built the data infrastructure that makes the analysis possible by digitizing the NRMP match data.

2 Background

Understanding how Medicaid expansion may have altered hospital training decisions requires engaging with three layers of institutional context. First, the expansion fundamentally changed the financial environment of teaching hospitals by converting uncompensated care into Medicaid revenue, but the magnitude and direction of that change varied across institutions depending on their pre-existing patient mix, their state's disproportionate share hospital (DSH) formula, and the design of state Medicaid graduate medical education (GME) funding. Second, residency training operates within a tightly regulated system of federal funding caps, accreditation requirements, and a centralized matching market, each of which shapes how hospitals translate financial incentives into slot decisions. Third, teaching hospitals are not a homogeneous group: they differ systematically in Medicaid exposure, ownership structure, training intensity, and geographic setting in ways that generate predictable heterogeneity in their responses to the same policy. This section delves into each of these layers in turn.

2.1 The Affordable Care Act's Medicaid Expansion

Prior to the Affordable Care Act (ACA), approximately 35.6 million non-elderly adults in the United States lacked health insurance (Vistnes and Keenan 2019). Teaching and safety-net hospitals absorbed the resulting uncompensated care costs through commercial cross-subsidization and federal DSH payments, effectively operating as insurers of last resort at an estimated cost of roughly $800 per uninsured individual annually (Garthwaite, Gross, and Notowidigdo 2018; Lewin and Altman 2000).

The ACA extended Medicaid eligibility to nearly all adults with incomes at or below 138 percent of the federal poverty level. Following *National Federation of Independent Business v. Sebelius*, 567 U.S. 519 (2012), adoption became voluntary at the state-level, producing the staggered rollout that underlies the identification strategy in this paper.[1] Twenty-five states and the District of Columbia expanded effective January 2014; additional states followed later in 2014 and in 2015, 2016,

1. The Supreme Court held that conditioning all federal Medicaid funds on adoption of the expansion was unconstitutionally coercive (*NFIB* 2012). Expansion thus became a state option rather than a requirement, generating the staggered adoption exploited here.

and 2019, while seventeen had not expanded as of 2019 (KFF 2026).[2] The federal government financed the newly eligible population at 100 percent through 2016, phasing down to a permanent 90 percent match by 2020, well above the standard federal medical assistance percentage (FMAP) of 57 to 63 percent. A large literature documents the resulting gains in coverage (Courtemanche et al. 2016; Glied and Jackson 2017; Kaestner et al. 2017), health outcomes (Courtemanche et al. 2018; Miller, Johnson, and Wherry 2021; Sommers et al. 2015; Sommers et al. 2017), and healthcare utilization (Courtemanche et al. 2017; Kobayashi et al. 2019; Nikpay et al. 2017). These studies focus on demand-side outcomes; the supply-side consequences for physician training capacity are the focus of this paper.

The hospital-level financial consequences of expansion are well documented. Dranove, Garthwaite, and Ody (2016) find that uncompensated care fell from 4.1 to 3.1 percent of operating expenses at hospitals in expansion states, with the largest reductions at safety-net institutions. Blavin (2016) finds corresponding improvements in operating margins. These revenue gains, however, exist in direct tension with the ACA's concurrent reductions to DSH allotments, which are calculated as a function of uncompensated care volume and decline automatically as the uninsured share falls. Critically, following *NFIB v. Sebelius*, hospitals in non-expansion states faced mandated DSH cuts while continuing to absorb high uncompensated care volumes, creating a pronounced fiscal asymmetry (Linehan 2013; Ossei-Owusu 2017). The net revenue effect of expansion at any given hospital

2. Several states adopted expansion after 2019, but this paper's outcome data end in 2019, which captures the first five post-expansion years for early adopters; the panel begins in 2000, giving ten pre-expansion years.

therefore depends on its pre-expansion patient mix, its state's DSH formula, and the generosity of state Medicaid GME funding.

The ACA also incorporated targeted physician workforce provisions. Heisler (2013) records these, including Medicare payment adjustments, new GME funding streams, and the redistribution of unused Medicare residency slots under Section 5503 to hospitals in areas with documented physician shortages, with a statutory requirement that 75 percent of reallocated slots support primary care or general surgery training. McNamara and Hussain (2025) find that these cap changes increased program size, and McNamara and Pineda-Torres (2025) show that Section 5503 raised primary care physician supply by 5.2 percent in treated areas. These GME-targeted interventions confirm that the legislative environment surrounding Medicaid expansion was actively attempting to expand the training pipeline, making it essential to isolate any effect attributable to the Medicaid coverage channel specifically.

Never-expansion states also changed their GME policies during this period. Several of the largest ran state programs that operate through the same hospital-financing channel this paper studies, but without a coverage change. Florida created the Statewide Medicaid Residency Program effective July 2013. The program appropriates roughly $80 million per year in recurring state and federal funds and allocates payments to hospitals based on their shares of statewide residents and Medicaid inpatient reimbursement (Florida Office of Program Policy Analysis and Government Accountability 2023). Texas set a statutory target of 1.1 first-year residency positions per in-state medical graduate in 2011 and, beginning with a

$14 million appropriation in 2013, funded a growing series of GME expansion grants through its Higher Education Coordinating Board (Texas Higher Education Coordinating Board 2024). Georgia and Tennessee operated smaller programs of the same type, and Section 5503's redistribution of unused Medicare slots also favored hospitals in Southern non-expansion states (McNamara and Pineda-Torres 2025). These policies bias the institution-level comparison toward finding a contraction in expansion states: if control-state training grew because of these programs, part of any measured decline at existing institutions in expansion states reflects control-state growth rather than expansion-state contraction.

Evidence from earlier public insurance expansions establishes that coverage expansions can shift physician supply through institutional channels. Garthwaite (2012) shows that the Children's Health Insurance Program (CHIP) expansion changed physician behavior on both the participation and hours margins—the closest the closest point of comparison for our question of how a coverage expansion for our question of how a coverage expansion—and Barnes et al. (2023) document that public insurance expansions raise labor demand in physician practices. Qian (2024) shows that 1980s Medicaid expansions for pregnant women increased OBGYN counts in low-income counties, and Huh and Lin (2024) find these effects were persistent and driven by early-career physicians. Baker and Royalty (2000) shows that Medicaid demand shocks were absorbed primarily through public hospitals and clinics rather than private practices, a pattern directly relevant here given that hospitals control residency slot capacity. For the ACA expansion itself, Neprash et al. (2021) find increases in Medicaid participation and payer-mix shifts in primary care with little change in physician labor supply, and Candon et al. (2018) document that appointment availability for Medicaid patients tracks fee levels—both underscoring that the accommodation of new coverage runs through institutions rather than individual physicians. Because Medicaid pays physicians well below Medicare in most states (Zuckerman, Skopec, and Liang 2021), the revenue consequences of expansion for teaching hospitals depend on volume as much as price, which is why the financing formulas described below matter.

2.2 The US Physician Training System

2.2.1 Residency Training and the Match

Residency programs provide specialty-specific supervised clinical training and are a prerequisite for board certification and independent licensure in all specialties. Programs are accredited by the Accreditation Council for Graduate Medical Education (ACGME) and vary in length from three years in family medicine, internal medicine, and pediatrics to five years in general surgery. Any constraint on residency capacity translates directly into a constraint on future physician supply, and where physicians train strongly predicts where they ultimately practice (LeFevre and Colwill 1983; Murphy 2020; Orr 2023), making hospital-level training decisions a important determinant of long-run physician workforce distribution. Entry into residency is coordinated through the National Resident Matching Program (NRMP) Main Residency Match, a centralized clearinghouse in which medical school graduates and programs submit rank-ordered preference lists that are resolved by a deferred acceptance algorithm (Gale and Shapley 1962; Roth and Peranson 1999); Agarwal (2015) estimates an empirical model of this market and shows that program capacities, not applicant scarcity, bind for most programs. Approximately 40,000 positions are offered annually across more than 6,000 programs.

2.2.2 Creating and Expanding Residency Programs

Developing a new residency program requires substantial institutional commitment well before the first resident class. Barajaz and Turner (2016) describe four overlapping phases: infrastructure development, educational design, ACGME accreditation, and recruitment. Programs must demonstrate sufficient physical, human, and financial resources, and ACGME approval is not automatic (Accreditation Council for Graduate Medical Education 2022). These preparatory investments represent sunk costs committed years in advance, making new program creation unlikely in response to anything other than a durable change in the financial environment.

Expansion of existing programs is constrained by the Balanced Budget Act of 1997, which fixed each hospital's Medicare-funded resident count at its 1996 level. Training residents above this cap requires hospitals to finance the full marginal cost from operating budgets, covering salaries, benefits, and faculty supervision time. Medicaid GME add-on payments partially offset this cost in states with volume-responsive formulas, but above-cap expansion is generally a discretionary expenditure sensitive to operating margins. Program complements introduce a further constraint: each program's maximum resident count is set by the ACGME based on faculty availability and clinical caseload, and increasing a complement requires formal approval (Accreditation Council for Graduate Medical Education 2022). Financial capacity and faculty availability jointly determine whether a hospital can meet the conditions for growth even when the regulatory pathway exists.

2.2.3 Graduate Medical Education (GME) Funding

GME in the United States is financed through a multi-payer system including Medicare, Medicaid, the Health Resources and Services Administration (HRSA), the Veterans Health Administration (VHA), and the Department of Defense (DoD) (He, Whang, and Kristo 2021). Medicare is the dominant source, providing Direct (DGME) and Indirect Medical Education (IME) payments totaling approximately $16.2 billion in fiscal year 2020 (Congressional Research Service 2022); whether these subsidies determine the level of training, or instead accrue to hospitals training residents they would fund anyway, is a long-standing question (Newhouse and Wilensky 2001; Nicholson 2002), and Medicare Payment Advisory Commission (2023) reviews the current policy debate. Its payment *formula* is fixed in federal statute, so Medicaid expansion does not change Medicare GME rules directly. Realized Medicare payments can nonetheless move with a hospital's resident counts and the patient-share terms of the formula, which is why we treat measured payments as an equilibrium outcome rather than a mechanical transfer when we use them as evidence on the financing channel. Medicaid GME ranks second at $7.39 billion in 2022, up from $3.78 billion in 2009 (Henderson 2021, 2023), but unlike Medicare it is entirely state-determined. States vary substantially in whether they fund GME at all, in payment generosity, and in formula design (Eden, Berwick, and Wilensky 2014; Henderson 2016, 2021, 2023; National Conference of State Legislatures 2024). Some operate volume-responsive systems that scale payments with Medicaid patient volume; others use fixed per-resident amounts, managed care capitation rates, or direct grants. Federal matching amplifies all state payments, so state-level formula design choices have outsized consequences for teaching hospital resources. In states with volume-responsive formulas, expansion mechanically raises Medicaid patient volume and triggers GME revenue increases without additional legislation. In states with fixed formulas, the same volume gain generates no incremental GME revenue. This cross-state variation is a key source of identifying variation in the mechanism analysis. HRSA's Children's Hospital GME and Teaching Health Center GME programs provide additional targeted support for pediatric and primary care training, insulating those programs from the operating margin pressures that affect procedurally intensive specialties (He, Whang, and Kristo 2021).

3 Conceptual Framework

Consider a simple framework for how residency positions are determined. Residency training has two sides: hospitals decide how many positions to create and fund, and medical graduates apply to fill them. The number of residents a hospital actually trains is the smaller of the two—the positions it offers, or the applicants willing to take them.

In practice, the offering side is what usually binds. In our data, the large majority of programs—82 percent overall, 80 percent in expansion states and 87 percent in never-expansion states—fill every position they offer (Table 1). For the typical program, then, the number of residents trained is set by how many positions the hospital is willing and able to fund, not by a shortage of applicants; the roughly one program in five with unfilled positions is a real margin, but a minority one. The question that matters is therefore what governs a hospital's decision to fund positions in the first place.

That decision is, above all, financial. Medicare pays for residents only up to a cap set in 1997; beyond that cap, a hospital must cover the full cost of each additional position—salary, benefits, and faculty supervision—out of its own operating budget. The marginal position is thus a discretionary expense that a hospital will sustain only if its finances allow, propped up by a mix of Medicare and Medicaid education payments and the disproportionate-share and uncompensated-care funds that help keep teaching hospitals afloat.

The same financial calculus operates on a second margin the literature usually ignores: *entry*. An institution that does not yet train residents faces the accreditation process and start-up costs so the entry decision responds to durable changes in the financial environment, not transitory ones. A policy that changes hospital finances can therefore move training capacity in two distinct ways: existing sponsors adjust the positions they fund, and new sponsors decide whether the now-different environment justifies starting a program. Nothing in the framework requires the two margins to move together; a shock could leave intake at existing institutions unchanged while accelerating (or deterring) entry, in which case aggregate capacity changes would appear only in totals that include entrants, and hospital panels restricted to existing institutions would miss them entirely.

Medicaid expansion changes this financial picture, but not in a single direction. On one hand, it turns previously uninsured patients into paying—if low-margin—Medicaid patients, which improves revenue. On the other, it reduces the uncompensated-care payments that are tied to the size of the uninsured population, which cuts the other way. Whether a given hospital ends up with more or less room to fund positions depends on its patient mix and on how its state pays for graduate medical education. The framework therefore delivers one clear prediction—that Medicaid expansion changes residency training by changing hospital finances—while leaving both the *direction* of the change and the *margin* on which it operates (intake at existing institutions versus entry) to be settled empirically.

4 Data

We use data from the NRMP Main Residency Match covering 2000 to 2019. The 2010 to 2019 years come from the NRMP's published data reports (National Resident Matching Program 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019). The reports document, at the program level, positions offered, positions filled, match outcomes, and fill rates, disaggregated by specialty. We identify the sponsoring institution for each program using the leading digits of the NRMP program code, which correspond to a unique hospital or training institution. Aggregating programs to the institution level yields a panel of 1,234 sponsoring institutions over twenty years. Panel construction handles the extensive margin explicitly: an institution-year in which the sponsor does not appear in the NRMP report is recorded as *missing*, not zero, outside the window between the institution's first and last observed activity, so entry and exit are never conflated with capacity changes at existing institutions.

Entry into the NRMP requires careful measurement because of the osteopathic single-accreditation transition. Between July 2015 and June 2020, residency programs accredited by the American Osteopathic Association (AOA) moved to ACGME accreditation, and their sponsoring institutions entered the NRMP Main Match. These institutions trained residents before the transition; their appearance in match data reflects a change in accreditation rather than new training capacity. We therefore classify each of the 201 institutions that first appear in the match after 2010 as either a genuinely new sponsor or a pre-existing AOA-accredited sponsor. The classification matches entrant institutions, within state, against the ACGME Accreditation Data System's public list of all 743 programs that applied for accreditation under the Single Accreditation System, including programs that later withdrew. Of the 201 post-2010 entrants, 100 are single-accreditation migrants and 101 are genuinely new sponsors.

Specialty-level analyses use the same panel aggregated to institution-by-specialty-group cells. Because specialty-specific counts are conditional on a hospital sponsoring a program in that specialty, the pre-expansion means and per-program magnitudes reported by specialty and by hospital location need not sum to the all-specialty aggregate.

The primary outcome is matched positions per 100,000 population: the total number of positions filled at a given hospital in a given year, divided by the state population and scaled to a per-100,000 basis. Population data are from the U.S. Census Bureau (U.S. Census Bureau 2000, 2019). We analyze both the aggregate total across all specialties and specialty-specific counts. Table 1 reports baseline (2010) summary statistics by expansion status: expansion and never-expansion states host programs of similar size, fill rates, rural composition, and specialty mix, though expansion states train substantially more physicians per capita—a level difference the program fixed effects absorb.

When the NRMP lists multiple specialties for a single program (e.g., "Emergency Medicine/ Anesthesiology"), slots are assigned to the first-listed specialty prior to aggregation to the hospital level. Because specialty models are estimated independently, this convention affects only the first-listed specialty's counts; the second specialty's count is unchanged. As a robustness check, we also allocate a slot to each listed specialty, which increases counts only for second-listed specialties.

A second dataset measures the financial side of training. From the Centers for Medicare and Medicaid Services' annual Graduate Medical Education files—extracted from the Medicare hospital cost reports—we assemble a hospital-by-fiscal-year panel of 1,933 teaching hospitals covering fiscal years 2000 through 2023, identified by Medicare provider number. The files report each hospital's Direct GME and Indirect Medical Education payments, resident full-time-equivalent counts, and resident caps. Because cost-reporting periods do not always span twelve months, we annualize all payment and FTE amounts by months covered and sum multiple reports filed by the same provider within a fiscal year. These are *Medicare* payments—a proxy for training-related revenue rather than a direct measure of the state Medicaid GME payments at the center of the financing mechanism—and we use them for the payment event studies and, through a crosswalk from NRMP sponsoring institutions to Medicare provider numbers (683 linked institutions), for the analyses that measure match intake and cost-report resident stocks on the same institutions.

Figure 1 summarizes the treatment structure of the panel, displaying each state's expansion status in every sample year. Treatment is the host state's adoption of Medicaid expansion. States expanded in distinct waves: twenty-six states and the District of Columbia expanded in 2014, and later adopters followed in 2015, 2016, and 2019. Seventeen states never expanded over the sample period, and hospitals in those states serve as the comparison group. This staggered rollout is the source of identifying variation for the event-study design.

Figures 2 and 3 plot the raw evolution of matched positions by expansion cohort, in totals and on a per-100,000 basis, where a cohort is defined by the calendar year in which a hospital's state expanded Medicaid. These series are purely descriptive and are not adjusted for fixed effects or covariates; they show that matched positions trend upward across all cohorts over the sample period, with treated and never-expansion hospitals moving broadly in parallel prior to expansion.

Finally, Figure 4 situates residency positions within the aggregate physician workforce. Using the American Community Survey, it plots the cumulative growth since 2001 in physicians per 100,000 population against growth in total population, each indexed to its 2001 value. Over nearly two decades physicians per 100,000 have risen at roughly the same pace as the population itself. Because the long-run supply of physicians is shaped upstream by the number of residency positions, this near-parallel growth motivates our focus on graduate medical education as the margin through which shocks to hospital finances may propagate into the physician workforce.

5 Empirical Strategy

In this paper, we estimate the dynamic effects of Medicaid expansion on the number of matched residency spots at a hospital per 100,000 state population using the imputation estimator developed by Borusyak, Jaravel, and Spiess (2024). This approach addresses the biases that arise when using conventional two-way fixed effects (TWFE) estimators in settings with staggered adoption difference-in-differences designs (De Chaisemartin and d'Haultfoeuille 2020, 2023; Goodman-Bacon 2021; Roth et al. 2023; Sun and Abraham 2021). We now discuss the model, identification assumptions, and estimation approach.

Let Y_{ist} denote the number of matched residency spots per 100,000 state population at hospital i in state s, and year t. Following Borusyak, Jaravel, and Spiess

(2024), we specify the following event study model that allows for unrestricted treatment effect heterogeneity:

$$Y_{ist} = \lambda_i + \gamma_t + D_{ist}\tau_{ist} + \varepsilon_{ist} \tag{1}$$

where λ_i represents hospital fixed effects, and γ_t are time fixed effects. D_{ist} is an indicator equal to one if Medicaid was expanded in state s at time t, and τ_{ist} represents the fully heterogeneous treatment effect for hospital i at time t. In equation (1), for each hospital, the data contains an activation date, E_i, when D_{ist} switches from 0 to 1. This specification allows treatment effects to vary arbitrarily across hospitals and time periods without imposing parametric restrictions.

Our identification strategy leverages the staggered roll-out of Medicaid expansion across states between 2014 and 2019. The model in equation (1) is generated from three main assumptions on potential outcomes and causal effects. First, the parallel trends assumption requires that, absent Medicaid expansion, the number of matched residency spots per 100,000 state population would have evolved similarly between hospitals; because ours is an event-study design, we assess this assumption directly by testing for differential pre-expansion trends below. Second, we assume no anticipation—that Medicaid expansion did not affect the number of matched spots per 100,000 state population before the law's actual implementation in each state. This assumption is plausible in our setting: expansion timing was set by state-level political processes following *NFIB v. Sebelius* rather than by the training decisions of individual hospitals, and the multi-year lead times required to create or expand a residency program make preemptive adjustment unlikely.

A further assumption implicit in equation (1) is that potential outcomes are unaffected by other states' treatment—the stable unit treatment value assumption (SUTVA). It deserves explicit discussion here because residency positions are filled through a single national clearinghouse: if expansion leads treated-state programs to offer fewer positions, some applicants who would have matched there may instead match at programs in non-expansion states, mechanically raising outcomes in the comparison group. Spillovers of this form would inflate the treated–control gap, so our estimates should be read as an upper bound (in magnitude) on the per-state effect of expansion. Two features of the setting limit the scope for such interference. First, the relevant absorption margin is slack in the *comparison* group, where 87 percent of programs fill every position they offer (Table 1; the expansion-state rate is 80 percent), so comparison-state programs can absorb displaced applicants only where slack exists. Second, and more important, the number of positions a program *offers* (its quota) is set by hospital funding decisions before the match algorithm runs and cannot be inflated by the equilibrium reallocation of applicants.

Finally, we impose a model of unrestricted causal effects, referred to as the "null model" in Borusyak, Jaravel, and Spiess (2024). In this case, the target estimand (parameter of interest) is the dynamic average treatment effect on the treated (ATT) h periods (horizons) since the treatment for a given $h \geqslant 0$:

$$\tau_h = \sum_{\{i,s,t\}:K_{ist}=h} w_{ist}\tau_{ist} \tag{2}$$

where $K_{ist} = t - E_i$ denotes event time—the number of years elapsed since Medicaid expansion in hospital i's state—and the weight is given by $w_{ist} = \frac{\mathbb{1}(K_{ist}=h)}{|\{i,s,t\}:K_{ist}=h|}$ and sums to one within each event time h. Borusyak, Jaravel, and Spiess (2024) proposes an imputation estimator that uses untreated observations to predict what would have happened to treated units in the absence of treatment.

The estimator proceeds in three steps:

1. Using only the untreated units (i.e., observations with $D_{ist} = 0$) and ordinary least squares (OLS), we obtain $\hat{\lambda}_i$, $\hat{\gamma}_t$, and $\hat{\delta}$ from

$$Y_{ist} = \lambda_i + \gamma_t + X'_{ist}\delta + \varepsilon_{ist},$$

   where X_{ist} is a vector of predetermined covariates. In our baseline specification X_{ist} is empty—the counterfactual is built from the hospital and year fixed effects alone—and we use it only in robustness checks (e.g., adding baseline state population as a control).

2. For each treated observation $\{i, s, t\}$ with $D_{ist} = 1$, we construct the untreated potential outcome (counterfactual outcome) as $\hat{Y}_{ist}(0) = \hat{\lambda}_i + \hat{\gamma}_t + X'_{ist}\hat{\delta}$ and estimate the individual-specific treatment effect as $\hat{\tau}_{ist} = Y_{ist} - \hat{Y}_{ist}(0)$.

3. Estimate the event-time coefficients as weighted averages: $\hat{\tau}_h = \sum_{\{i,s,t\}:K_{ist}=h} w_{ist}\hat{\tau}_{ist}$.

We cluster standard errors at the state level to account for potential serial correlation within states over time that are robust to arbitrary forms of heteroskedasticity. Although the maintained assumptions of the difference-in-differences design are untestable in the post-treatment period, we can perform a robust test of the identifying assumptions in the pre-treatment period (pre-trends test). Unlike the conventional pre-trends test using standard event studies, the imputation-based method affords the opportunity to test for parallel pre-trends and no-anticipation assumptions using only the untreated observations.

To conduct the pre-trends test, one needs to choose an alternative model for the outcome Y_{ist} for the untreated observations. Specifically, for an observable vector W_{ist}, the alternative model may be written as $Y_{ist} = \lambda_i + \gamma_t + X'_{ist}\delta + W_{ist}\theta + \varepsilon_{ist}$, where W_{ist} may represent a set of binary indicators for $1, \ldots, k$ periods prior to the start of the treatment for some chosen k. Next, using the untreated observations only, obtain the OLS estimate of θ and test the hypothesis $\theta = 0$. We present all our main results graphically, combining these pre-trend estimates with the horizon-specific ATTs from equation (2). As discussed in Borusyak, Jaravel, and Spiess (2024), this robust OLS-based pre-trends test avoids the pre-testing concerns in Roth (2022). Specifically, regression-based tests use the full sample, including the treated observations, thereby imposing restrictions on treatment effect heterogeneity. Moreover, conducting inference using the imputation estimates of the ATT remains valid even if we condition on passing the pre-trends, avoiding the issue of inflated variances and overly conservative inference that often arises with standard pre-trend tests (Roth 2022).

6 Results

6.1 Main Effect of Medicaid Expansion on Residency Positions

Figure 5 plots the dynamic treatment effects from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), with the outcome measured as matched positions per 100,000 contemporary state population at existing institutions.[3] The average post-expansion effect is -0.014 positions per 100,000, about 5.7 percent of the pre-expansion mean of 0.26. The estimate is statistically insignificant. The post-expansion coefficients are negative and grow more negative over the time (joint test $p = 0.05$). The ten pre-expansion coefficients are jointly flat (pre-trend $p = 0.36$), supporting the parallel trends assumption. Offered positions—the quota programs post before the matching with medical students—barely move: -0.008 per 100,000, about 3.0 percent, statistically insignificant ($p = 0.21$) with flat pre-trends ($p = 0.39$). Hospitals did not stop posting positions after expansion. Although matched positions at existing institutions experience a slight drop, the capacity-creation margin in our conceptual framework remains entirely unaffected.

We scale by contemporary rather than fixed population deliberately. Normalizing matched positions by year-varying state population nets differential population growth out of the outcome; the raw count of positions per institution, by contrast, exhibits a pronounced differential pre-trend (Appendix Figure A.1) that confounds a levels interpretation, so we treat levels as evidence on the direction of the effect

3. Outcomes outside an institution's activity window—the span from its first to its last year with positive positions—are coded as missing rather than zero, so identification comes only from changes at operating institutions.

only.

6.2 The Extensive Margin: Entry, Exit, and the Single Accreditation System

The question this subsection answers is: whether expansion induced entry. Of the 1,234 sponsoring institutions in the panel, 201 first appear in the match after 2010—156 of them from 2016 onward, the window of the osteopathic (AOA-to-ACGME) single-accreditation transition. The intake-margin estimates are insensitive to these entrants (Table 2): restricting to institutions active over the entire panel gives -5.5 percent, excluding all 100 classified single-accreditation migrants gives -5.7 percent (pre-trend $p = 0.38$), and excluding every 2016-and-later entrant gives -5.2 percent, all within a standard error of the headline and statistically indistinguishable from zero.

Classifying the entrants changes the interpretation of the entry margin (Table 2). By 2019, post-2010 entrants account for 2,148 matched positions in expansion states, but 982 of those positions (46 percent) are at single-accreditation migrants—institutions that trained residents under AOA accreditation before 2015 and became visible in match data when their accreditation changed. Among expansion-state entrants in the 2016–2019 window, 63 percent are migrants, because osteopathic training was historically concentrated in Michigan, Ohio, Pennsylvania, New York, and New Jersey, all of which expanded Medicaid. Genuine-entrant capacity per 100,000 shows no response to expansion: the population-weighted average post-expansion effect is -0.12 (joint $p = 0.19$) and the unweighted effect is $+0.02$ (joint

$p = 0.34$), both have statistically insignificant pre-trends (Appendix Figure A.2). Migrant capacity, by contrast, rises strongly in the post period (joint $p < 0.001$)—not a treatment effect, but the national accreditation calendar interacting with the geography of osteopathic training.

At the state level, where entry belongs in the total, per-capita training does not fall after expansion. The point estimate is positive (+6.7 percent; Appendix Figure A.3) but fails its state-level pre-trends test ($p = 0.0002$). The unweighted estimate is +4.6 percent with $p = 0.38$, and it is the only entry-inclusive specification in the paper with statistically insignificant pre-trends test ($p = 0.17$). Excluding migrant capacity changes little: the weighted estimate is +6.2 percent and the unweighted estimate is +4.1 percent (pre-trend $p = 0.16$; Appendix Figure A.4), so the non-declining state total is not an artifact of the accreditation transition. The state-level evidence supports our conclusion that totals did not decline. Moreover, most of the treated-control difference in entry reflects reclassification rather than new capacity.

6.3 Mechanism: The Medicaid GME Financing Channel

The framework leaves the sign of the demand shift to the data because expansion changes hospital finances in offsetting ways: it converts uncompensated care into billable Medicaid revenue and, in states whose Medicaid GME formulas scale with patient volume, raises GME add-on payments, while the Medicare uncompensated-care pool and below-cost Medicaid reimbursement push the other way. We isolate the operative channel using cross-state variation in Medicaid

GME formula design: *volume-responsive* states gain GME revenue when expansion raises Medicaid volume, whereas *fixed* and non-paying states gain none. States are classified by the payment rules in force during the 2014–2019 treatment window, from the survey of Henderson (2016); an exposure-weighted classification that follows each state's rules year by year assigns every expansion state the same arm. Because the rules are measured partly within the treatment window—New York and Illinois, among others, adopted volume-scaled formulas by 2015—the classification is a policy choice rather than a pre-determined trait. If the financing channel operates, the contraction should be concentrated in the fixed/non-paying group.

The first stage of the financing channel is a revenue shock: for expansion to move training through hospital finances, it must first move the GME payments hospitals receive. Figures 6 and 7 test this with event studies of Direct GME and Indirect Medical Education payments from the Medicare cost reports. The inverse-hyperbolic-sine specification suggests a first stage exists—Direct GME payments rise roughly 15 percent (joint $p < 0.01$; Figure 6), with a smaller IME response (Figure 7)—but the result is not robust to functional form (Chen and Roth 2024). Decomposed into intensive and extensive margins, the response vanishes: on hospital-years with positive payments (98 percent of the sample for Direct GME), the log specification gives -4.3 percent for Direct GME ($p = 0.57$) and -5.7 percent for IME ($p = 0.55$), and the probability that a hospital receives any payment is unchanged (Appendix Table A.1). Expansion did not generate a detectable revenue shock.

Figure 8 shows the cross-formula comparison. In non-responsive (fixed/none) expansion states, the average post-expansion effect is -0.035 per 100,000—a 17.8 percent decline relative to the pre-expansion mean of 0.19—following pre-expansion coefficients that are flat over all ten pre-periods (pre-trend $p = 0.78$). In volume-responsive states the average post-expansion effect is $+0.012$ per 100,000 ($+3.7$ percent relative to a pre-expansion mean of 0.33), but that panel fails its ten-period pre-trends test ($p < 0.001$), so its level is not interpretable as a treatment effect. A pooled interacted model puts the volume-minus-non-responsive difference in average post effects at 0.049 per 100,000 ($p < 0.001$), robust to flipping each of the five states whose coding required judgment but with $p = 0.53$ under wild cluster bootstrap-t inference (Appendix Table A.2).

Two caveats bound this comparison, and both attach to the volume-responsive arm rather than to the design as a whole. First, the volume arm fails its ten-period pre-trends test, so its level—and the cross-arm difference that inherits it—should be read descriptively; the contraction in non-responsive states follows ten flat pre-expansion coefficients and stands on its own. Second, the formula classification is a state policy choice, not a randomized assignment (the coding, including the California case, is documented in Appendix appendix C). Appendix Table A.3 shows the two groups of expansion states are balanced on baseline observables with identical expansion timing—the population-*weighted* program-level baseline (roughly 0.35 versus 0.20 matched positions per 100,000) reflects the concentration of training in a few large volume-responsive states rather than state-level imbalance—but a state that expected expansion to raise Medicaid volume could have adopted a volume-scaled formula for that reason, and a cross-state split cannot rule that out.

The decisive test is the payment response itself: if the mechanism operates, the GME revenue gain from expansion should be concentrated in volume-responsive states. Appendix Figures A.5 and A.6 re-estimate the payment event studies separately by formula design, and no cross-arm gradient appears: the DGME difference is -0.02 ($p = 0.87$), the IME difference -0.04 ($p = 0.56$), and the resident-FTE difference -0.01 ($p = 0.86$). The revenue gain the mechanism requires is not concentrated in volume-responsive states; it is not detectable at all.

We conclude that the data do not establish a Medicaid GME financing mechanism. The cross-arm difference is driven by the contraction in non-responsive states rather than by growth in volume-responsive ones, is indistinguishable from zero under wild cluster bootstrap-t and randomization inference, and is unaccompanied by the revenue shock the channel requires. What the exercise establishes is narrower and still useful: the intake decline is concentrated in states whose formulas do not reward Medicaid volume—a pattern consistent with the financing channel's prediction but, absent a payment response, not attributable to it. Future work splitting states on these classifications should report the judgment-flip and log-level versions of its results.

6.4 Heterogeneous Effects by Hospital Location

The intake-margin estimates mask heterogeneity across hospital settings, and on the full panel the heterogeneity runs opposite to the shortage-geography worry. Figure 9 estimates the effect separately for urban and rural institutions in two *split-sample* regressions: each group is compared only with never-expansion institutions of the same type, and each carries its own ten-period pre-trends test. (The 2020 Rural-Urban Commuting Area (RUCA) classifications are unavailable for institutions that exited before 2010 and for Alaska and Hawaii; those institutions are excluded from this split only.) For urban institutions, which account for the large majority of training volume, the average post-expansion effect is -0.014 per 100,000 (s.e. 0.027; -5.0 percent; joint $p = 0.03$), though the urban pre-trends test now rejects at the ten-period horizon ($p = 0.02$), so the urban path should be read with the same caution as the levels evidence. For rural institutions the estimate is *zero* to three decimal places ($+0.0001$ per 100,000; $+0.1$ percent), against a rural baseline of 0.06 per 100,000—rural training at existing institutions simply did not move with expansion, though the rural pre-trends test also rejects ($p = 0.009$). A pooled interacted model puts the urban-minus-rural difference at -0.045 ($p < 0.001$), but because both split samples fail their own pre-trends tests on the long window, we read the location split as descriptive: nothing in it suggests expansion drained rural training, and the action, such as it is, sits at large urban institutions.

6.5 Heterogeneous Effects by Specialty

We next examine whether the intake-margin effect differs across specialties, distinguishing primary care (family medicine, internal medicine, and pediatrics) from all other specialties, estimated at the institution-by-specialty-group level so that each group carries its own unit fixed effects. Figure 10 presents the estimates. Primary-care intake at existing institutions is essentially flat: -0.001 per 100,000, a

0.9 percent change (s.e. 0.014; joint $p = 0.06$; pre-trend $p = 0.04$). The non-primary-care estimate is larger but imprecise: -0.019 per 100,000, an 11.1 percent decline (s.e. 0.017; joint $p = 0.14$; pre-trend $p = 0.01$). Both panels' pre-trends tests reject on the ten-period window, so we read the specialty split as suggestive. Its practical content is reassuring rather than alarming: the specialty most tied to the shortage debate—primary care—shows no intake contraction at all, and the group estimates approximately aggregate to the total.

7 Robustness Checks

7.1 Alternative Difference-in-Differences Estimators

To demonstrate that our findings are not an artifact of the particular estimator we adopt, we re-estimate the event study using a suite of recently developed difference-in-differences estimators that are robust to treatment-effect heterogeneity under staggered adoption. Figure 11 compares six approaches for matched positions per 100,000 contemporary population: two-way fixed effects (TWFE), Borusyak, Jaravel, and Spiess (2024), De Chaisemartin and d'Haultfoeuille (2020), Sun and Abraham (2021), Cengiz et al. (2019), and Gardner (2022). Each addresses the "forbidden comparisons" that can contaminate conventional TWFE estimates under staggered timing through a different methodological route. Across estimators, the post-expansion estimates align closely in both sign and magnitude, and the pre-expansion coefficients are near zero throughout.[4] The same robustness holds for the levels outcome (Appendix Figure A.9) and the quota outcome (Appendix Figure A.10).

7.2 Unweighted Specifications

Our baseline specifications weight observations by state population so that the estimates summarize the experience of the average resident rather than the average institution. Completing the full weighting-by-denominator grid on the same panel: the weighted year-varying estimate is -0.014 (-5.7 percent; joint $p = 0.05$; pre-trend $p = 0.36$); the unweighted year-varying estimate is -0.018 (-3.1 percent of a much larger unweighted baseline; joint $p = 0.27$; pre-trend $p = 0.37$); the weighted fixed-2010-denominator estimate is -0.047 (-18.9 percent; joint $p = 0.03$) but its pre-trends test rejects ($p = 0.04$); and the unweighted fixed-denominator estimate is -0.026 (-4.6 percent; joint $p = 0.18$; pre-trend $p = 0.26$). The unweighted point estimates are not attenuated in absolute value; they sit on larger baselines and wider standard errors. A state-level collapse (51 units) yields a *positive* weighted estimate (+6.7 percent, $p = 0.10$, with failing state-level pre-trends) and a positive unweighted estimate (+4.6 percent, $p = 0.38$, pre-trend $p = 0.17$—the only entry-inclusive cell of the grid that passes its pre-trends test). No cell of the grid supports a precisely estimated intake contraction, and the only specifications with positive

4. The Callaway and Sant'Anna (2021) estimator is omitted from the figure for two reasons: it does not accept the population weights our design uses, and its never-treated comparison cells are sparsely populated in this sample. Estimated unweighted with the not-yet-treated control group, it gives a simple ATT of -0.005 per 100,000 (s.e. 0.007, $p = 0.46$)—a null fully consistent with the intake-margin reading. Differences across estimators can arise from the choice of weights used to aggregate group-time effects into an overall ATT; see, e.g., Deb et al. (2025).

point estimates are the ones that include entry. As Section 6.2 shows, part of that entry reflects single-accreditation migration rather than induced entry.

7.3 Pre-ACA Placebo

The check that does the most to rule out slow-moving regional confounds is a placebo that assigns each state an expansion ten years before its real one and re-estimates on 2000–2013 data alone, so no real treatment enters the sample (Appendix Figure A.11). The placebo comes back null at both the institution level ($p = 0.54$, flat placebo pre-trends) and the state level ($p = 0.32$): whatever separates expansion from non-expansion states after 2014 was not already separating them in the decade before the ACA.

7.4 Corroboration from the Linked Cost-Report Sample

The linked cost-report sample corroborates the intake-margin reading with an independent measure of the training stock. Matching NRMP institutions to Medicare provider numbers (683 linked institutions on the full panel) and estimating match intake and annualized cost-report resident FTEs on the same rows, weights, and year convention, the two series move together: both decline modestly (asinh intake −2.5 percent, asinh FTEs −1.2 percent), and neither arm of the formula split shows FTE growth on the linked sample (volume-responsive FTEs −2.5 percent, non-responsive −0.4 percent). The linked-subsample event studies fail their ten-period pre-trends tests, with p-values at or near zero in every cell, so we do not interpret these estimates causally. The reconciliation rests on the two series moving together on identical rows, not on either series being identified. Stock and flow tell one story once they are measured on the same institutions.

7.5 Inference with Few Clusters and Sensitivity to Parallel Trends

Because Medicaid expansion is assigned at the state level and the population-weighted design concentrates identifying variation in a small number of large expansion states, the effective number of clusters is well below the fifty-one states used to compute the clustered standard errors. Following the logic of Carter, Schnepel, and Steigerwald (2017) and approximating the effective number of clusters by $G^* = G/(1 + CV^2)$, the weighted design has $G^* \approx 8$ (the five largest states carry 66 percent of the weight), against $G^* \approx 22$ without weights. We therefore report several inference standards and are explicit about what each supports.

First, wild cluster bootstrap-t inference (Webb weights, state clusters, 9,999 replications) on the static two-way fixed-effects analogs of our estimates gives $p = 0.74$ for the headline intake margin and $p = 0.56$ for the timing-only design (Appendix Figure A.12)—fully consistent with the paper's reading that the intake margin is indistinguishable from zero. For the cross-formula mechanism difference the bootstrap gives $p = 0.53$.

Second, a Benjamini–Hochberg false-discovery-rate correction across the primary family of nine hypotheses—now including the two mechanism arms and the cross-arm difference—leaves no member of the family below $q = 0.05$ under clustered inference except the mechanism objects whose failing pre-trends disqualify them on other grounds (Appendix Table A.4 and Figure A.13).

Third, randomization inference: permuting the vector of state expansion-timing cohorts across states under the sharp null (1,000 draws for the headline, 500 for the other family members), in both the conventional form and the studentized form that permutes the t-statistic rather than the coefficient—the standard recommended when cluster weights are highly variable. The results are emphatic nulls: the headline has RI $p = 0.91$ (studentized 0.94), the timing-only design 0.65 (0.55), the cross-formula mechanism difference 0.48, and every other family member sits at 0.57 or above under either form. Nothing in the paper approaches conventional significance thresholds under randomization inference. For a paper whose central claim is a null on the intake margin, this conservative standard reinforces rather than threatens the conclusions.

Fourth, timing and influence diagnostics. Restricting treatment to the 2014 cohort alone holds cohort composition fixed and reproduces the headline (-0.014 per 100,000, -5.7 percent; joint $p = 0.02$) after ten flat pre-expansion coefficients ($p = 0.37$; Appendix Figure A.14). Dropping the 100 classified single-accreditation migrants leaves the headline essentially unchanged (-5.7 percent, pre-trend $p = 0.38$; Table 2). The genuine-entrant and migrant capacity estimates of Section 6.2 are reported under clustered inference only. They were constructed in response to review and are not members of the pre-specified nine-hypothesis family to which the randomization-inference and false-discovery-rate standards below apply. Excluding the four control states with the largest competing GME programs (Texas, Florida, Georgia, Tennessee) yields -0.083 (-32 percent, joint $p = 0.03$), but that specification fails its pre-trends test on the long window ($p < 0.001$), so we do not lean on it.

Fifth, the denominator (event studies for this block are collected in Appendix Figure A.15). Using log contemporary state population itself as the outcome shows the denominator trends differentially around expansion (pre-trend $p < 0.001$), in the direction that biases *against* finding a decline; a specification with log population as an estimated time-varying control absorbs most of the levels effect; and re-deflating by two alternative demographic denominators—the population aged 18–64 and the population below 150 percent of the federal poverty line (both available 2010–2019)—yields -3.7 and -5.6 percent respectively, each within a standard error of the headline. However, both alternative demographic deflators fail their ten-period pre-trends tests ($p < 0.001$). They corroborate the direction of the headline estimate but do not provide independent identification. Because every population-based denominator is potentially post-treatment, we also deflate by a non-demographic scale: current-dollar state GDP from the Bureau of Economic Analysis. This is the only alternative deflator that passes its ten-period pre-trends test ($p = 0.14$). The event-study path is flat on both sides of expansion, and the average post-expansion effect is $+3.9$ percent of baseline, with a confidence interval that comfortably includes zero. Across the four deflators we test, the two with flat pre-trends (contemporary population and state GDP) give estimates on opposite sides of zero, and the two that fail pre-trends give more negative estimates. This pattern is consistent with the intake-margin null: the dispersion across deflators reflects denominator trends rather than changes in training capacity. The raw levels outcome fails parallel trends outright (Appendix Figure A.1), as does the Rambachan and Roth (2023)-robust version of any levels claim, so levels serve only as directional corroboration. The Rambachan and Roth (2023) sensitivity analysis for the headline (Appendix Figure A.16) shows the robust confidence interval includes zero even at $\bar{M} = 0$—which, for a paper whose central claim is a null on this margin, is the expected result rather than a limitation.

Sixth, Medicare-cap exposure. Institutions training above their Medicare resident cap must fund marginal positions without federal DGME and IME support, so a financing channel operating through subsidized slots predicts a differential response by cap status. We find no such difference: the above-cap/below-cap difference in average post-expansion effects is 0.006 per 100,000 ($p = 0.88$). The test is underpowered, however. Only 42 institutions in 14 states train above their cap on the linked sample, and the above-cap arm fails its ten-period pre-trends test. We therefore read the comparison as descriptive: it fails to detect a difference rather than establishing its absence.

8 Policy Implications

What is at stake, in positions and dollars, is bounded by the estimates rather than asserted. At the intake margin, the effect of −5.7 percent carries a 95 percent confidence interval of [−0.066, +0.037] positions per 100,000—comfortably including zero in both directions; at the state level, where entry belongs in the total, the point estimates are positive (+6.7 percent weighted, +4.6 percent unweighted, neither statistically distinguishable from zero). Converting the most adverse plausible reading to dollars for perspective: public GME support runs roughly \$150,000 to \$180,000 per resident-year (Medicare DGME and IME totaled about \$16.2 billion; (Congressional Research Service 2022; Eden, Berwick, and Wilensky 2014)), so even a persistent loss of one thousand annual positions—well outside our state-level estimates—would represent roughly \$0.5–0.6 billion per year of forgone public training investment, against Medicaid GME spending of \$7.39 billion in 2022 (Henderson 2023). Because 50 to 60 percent of residents practice in their training state (Murphy 2020; Orr 2023), compositional shifts across institutions *within* states matter less for state physician supply than contraction would have; the incidence question our data can address is institutional (which sponsors train) rather than geographic.

Three implications follow. First, and most important to state plainly: nothing in this paper supports declining to expand Medicaid to protect the training pipeline. The corrected estimates show no aggregate contraction, the coverage literature documents large health and financial benefits of expansion (Miller, Johnson, and Wherry 2021; Sommers et al. 2017), and seven of the ten remaining non-expansion states have volume-responsive GME formulas—the configuration in which even the fragile cross-formula reading would predict no training loss. Second, expansion also did not *grow* the pipeline, and this result matters for workforce policy. Covering millions of additional people raises the demand for physicians, and the physician-shortage projections that motivate GME policy assume that training will expand to meet that demand. We find that the largest demand expansion in half a century produced no measurable supply response from existing institutions and no measurable entry by genuinely new sponsors. Coverage policy on its own does not finance training growth through hospital revenues. If the goal is to train more physicians for a larger insured population, the policies that moved training capacity over this period were direct interventions: Section 5503 redistribution (McNamara and Pineda-Torres 2025), Teaching Health Center GME, the CAA 2021 Section 126 slots, and the state programs. Third, for states weighing volume-responsive Medicaid GME formulas, the federal match makes conversion a comparatively cheap lever, but our evidence no longer supplies a demonstrated training return; the honest statement is that the formula margin remains untested.

9 Conclusion

How many physicians the United States trains—and in which specialties and places—is settled years upstream, in the residency positions that teaching hospitals choose to create and fund. Those decisions are, at bottom, financial, which leaves the training pipeline exposed to any policy that moves hospital revenue. This paper asks whether the ACA's Medicaid expansion, by reshaping the finances of teaching hospitals, altered the number of residency positions those hospitals offer and fill. Using the staggered rollout of expansion across states and the imputation estimator of Borusyak, Jaravel, and Spiess (2024), we find that training supply did not respond. Intake at existing institutions shows an imprecise decline that is indistinguishable from zero once entry and exit are coded as missing rather than zero, and offered positions barely move. We classify each post-2010 entrant against ACGME accreditation records and find that most of the apparent growth through newly entering institutions reflects the osteopathic single-accreditation transition, which moved pre-existing capacity into match data. Genuinely new sponsors show no differential entry response, and the migrating institutions, which are concentrated in expansion states because osteopathic training was historically concentrated there, account for most of the treated-control entry gap. State-level totals do not fall, with or without the migrating capacity. The offered-position margin behaves like the matched margin throughout, and the linked cost-report sample shows resident stocks tracking intake on comparable institutions. This decomposition reconciles the hospital-level and state-level evidence, including the companion estimates in Ang and Beniwal (2026): the two series include different amounts of migrating capacity, not different underlying training responses. A policymaker asking whether expansion shrank a state's training pipeline should use the state-level object; the answer is no. One asking whether expansion grew the pipeline should use the classified entry margin; the answer is also no. A placebo that assigns each state an expansion ten years before its real one, estimated on pre-ACA data alone, comes back null at both levels.

We also report, as a negative result, a test of the Medicaid GME financing channel. Classifying states by whether the formulas in force during the treatment window scale payments with patient volume, the decline is concentrated in states whose formulas do not; but the volume-responsive arm fails its ten-period pre-trends test, and payments estimated in logs on the positive subsample show no response at all. The formulas did not translate expansion into a detectable revenue shock, and the cross-formula contrast does not survive scrutiny.

Two features of the setting temper how precisely any of these magnitudes can be pinned down. First, treatment varies at the state level and the population-weighted design draws its identifying variation from a small number of large expansion states, so the effective number of clusters is limited; under randomization inference and wild cluster bootstrap, no individual estimate in the paper—intake margin, state total, or cross-formula difference—is statistically distinguishable from zero, and we say so wherever the estimate appears. Second, every panel construction carries its own pre-trend liability: the balanced panel is flat but zero-filled, the corrected panels are honest but fail pre-trends, and the contemporary-population denominator that delivers flat pre-trends is itself trending differentially (we show this directly by using log population as the outcome). Our confidence in the no-response reading rests on its *consistency across constructions with offsetting weaknesses*, not on any single clean design. Finally, our data capture positions offered and filled, not the downstream practice locations or health outcomes of the physicians those positions would have trained; linking training-capacity changes to eventual physician supply and population health is an important direction we leave to future work.

Taken together, the results show that the physician-training pipeline absorbed the largest coverage expansion in a generation without contracting and without growing. A demand expansion that insured millions of people, in an environment of projected physician shortages, produced no measurable supply response from the institutions that train physicians, on either the intake or the entry margin. Much of what looks like entry in post-2015 match data is an accreditation transition, and any study that uses NRMP or ACGME counts as an outcome needs to measure it deliberately. If training capacity is to keep pace with the insured population, the evidence from this period points to direct policy instruments rather than the hospital-finance channel. How coverage policy and training policy are coordinated, and who enters training provision under which accreditation and financing rules, is a natural direction for future research.

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Table 1: Summary Statistics at Baseline (2010)

Baseline characteristic (2010)	All states (1)	Expansion states (2)	Never-expansion (3)	Difference (2)−(3)	p-value
Matched positions per program	34.8	33.7	37.0	-3.3	0.562
Offered positions (quota) per program	36.7	35.7	38.8	-3.1	0.596
Share of programs filling every offered position	0.82	0.80	0.87	-0.07	0.130
Matched positions per 100,000 population	7.95	9.19	5.46	3.74	0.046
Offered positions (quota) per 100,000	8.36	9.66	5.76	3.90	0.042
Number of programs	12.9	14.9	8.9	6.0	0.088
State population (millions)	6.1	5.9	6.3	-0.4	0.834
Rural program share	0.07	0.06	0.08	-0.02	0.563
Primary-care share of matched positions	0.55	0.54	0.57	-0.03	0.554
Mean expansion year	2014.5	2014.5	–	–	–
Number of states	51	34	17		
Number of institutions active in 2010	658				
Institutions in panel (2000–2019)	1234				
Active institution-year observations (2000–2019)	13,892				

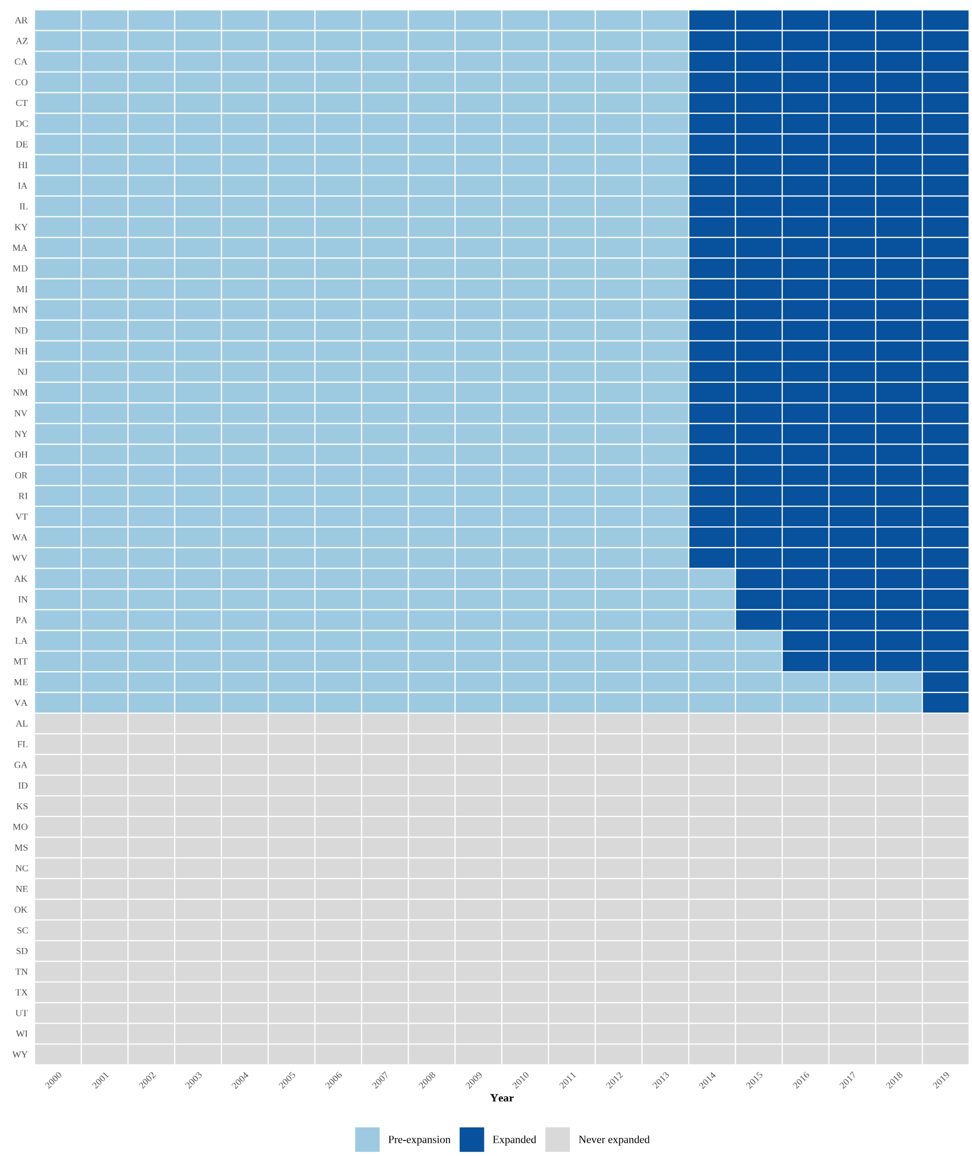
Notes: The table reports baseline (2010) means with the state as the unit of observation: each entry is the unweighted mean across states of the state-level value. Column (1) covers all 50 states and the District of Columbia; columns (2) and (3) split states by whether they ever adopted the ACA Medicaid expansion during 2010–2019; the p-value is from a Welch two-sample t-test of the expansion-versus-never difference. Program-level rows (matched and offered positions per program, rural share) first average across the state's programs, where a program is a sponsoring institution aggregated across specialties. Rural programs are those whose institution ZIP code (obtained by geocoding the institution address) carries a Rural-Urban Commuting Area (RUCA) code above 3 (RUCA unavailable for pre-2010 exiters and for Alaska and Hawaii institutions, which are excluded from the rural/urban split only); the primary-care share is the share of matched positions in family medicine, internal medicine, and pediatrics. Expansion states train more physicians per capita at baseline; the estimation design absorbs such level differences through program fixed effects. Data are from NRMP Main Residency Match reports and the U.S. Census Bureau.

Table 2: The Entry Margin Classified: Genuine Entrants versus Single-Accreditation Migrants

	Expansion states	Never-expansion states
Panel A: Post-2010 entrant institutions		
Entrants, 2011–2015		
Genuinely new sponsors	21	15
Single-accreditation migrants	5	4
Entrants, 2016–2019		
Genuinely new sponsors	42	23
Single-accreditation migrants	72	19
Matched positions at entrants, 2019		
At genuinely new sponsors	1,166	933
At single-accreditation migrants	982	426
Panel B: Event-study estimates		
	Average post effect	(Joint p; pre-trend p)
Genuine-entrant capacity per 100,000 (weighted)	−0.116 (0.154)	(p = 0.19; 0.19)
Genuine-entrant capacity per 100,000 (unweighted)	+0.025 (0.108)	(p = 0.34; 0.31)
Migrant capacity per 100,000 (weighted)	+0.047 (0.072)	(p < 0.001; 0.34)
State total excl. migrants (weighted)	+0.535 (0.380)	(p = 0.37; 0.0002)
State total excl. migrants (unweighted)	+0.364 (0.252)	(p = 0.40; 0.16)
Intake margin, always-active institutions (weighted)	−0.016 (0.029)	(p = 0.04; 0.45)
Intake margin, excl. migrants (weighted)	−0.015 (0.027)	(p = 0.08; 0.38)
Intake margin, excl. 2016-and-later entrants (weighted)	−0.009 (0.015)	(p = 0.17; 0.06)

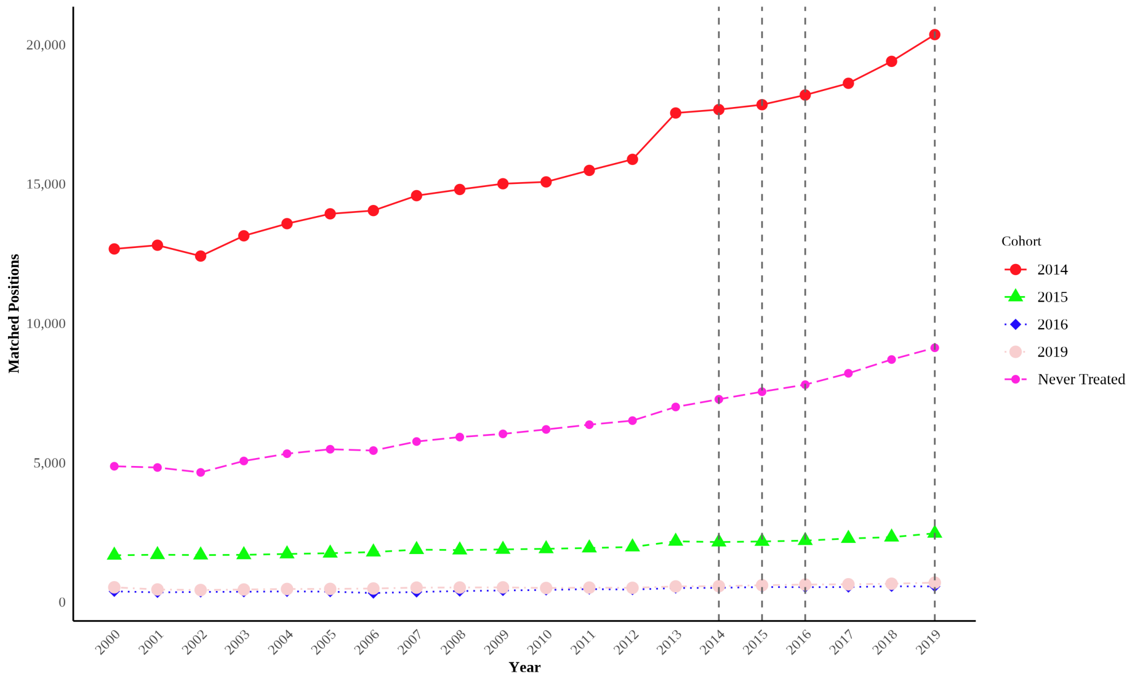
Notes: Panel A classifies the 201 institutions that first appear in the NRMP Main Match after 2010 as genuinely new sponsors or pre-existing American Osteopathic Association (AOA)-accredited sponsors migrating under the Single Accreditation System (July 2015–June 2020). Matched-position rows report total positions filled at those entrants in the 2019 match. Panel B reports average post-expansion effects from state-level event studies using the imputation estimator of Borusyak, Jaravel, and Spiess (2024) with state and year fixed effects, ten pre-expansion periods, and state-clustered standard errors (in parentheses); capacity outcomes are matched positions at the indicated entrant group per 100,000 contemporary population, so pre-expansion capacity at post-2010 entrants is near zero by construction and effects are reported in levels rather than percent. Weighted rows use 2010 state population weights. The migrant-capacity estimate reflects the national accreditation calendar interacting with the geography of osteopathic training, not a treatment effect. The three intake-margin rows re-estimate the institution-level headline specification on restricted samples: institutions active over the entire 2000–2019 panel, all institutions excluding the 100 classified migrants, and all institutions excluding every 2016-and-later entrant. Data: NRMP Main Residency Match reports; ACGME Accreditation Data System.

Figure 1: Staggered Adoption of Medicaid Expansion by State



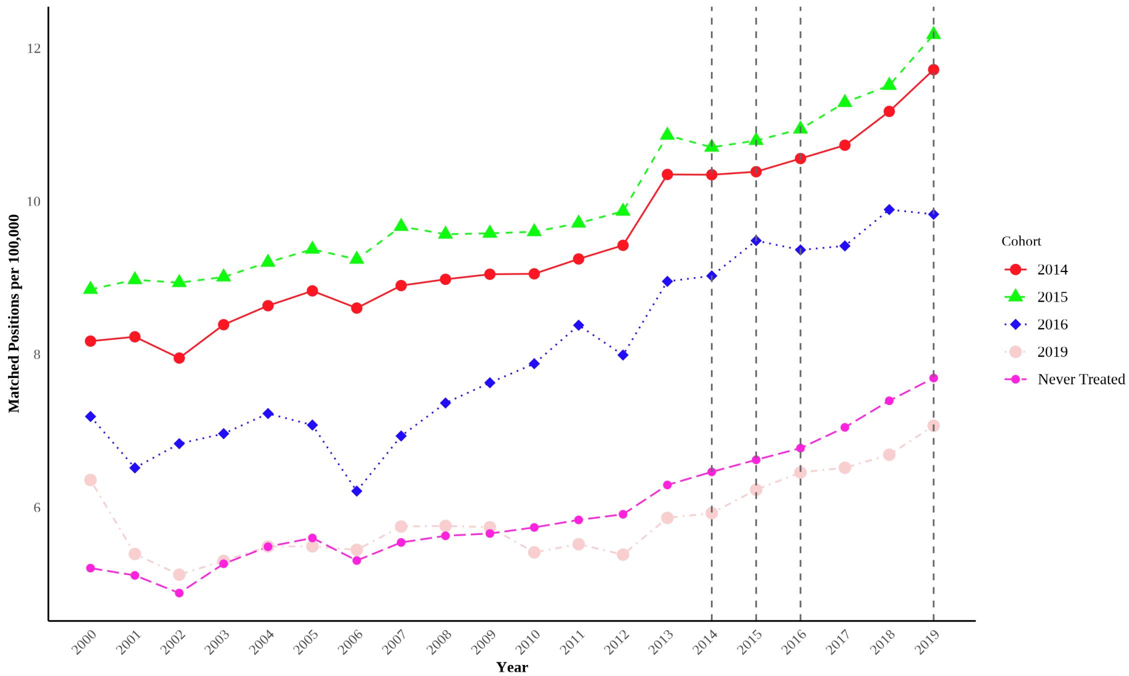
Notes: The figure shows the treatment status of every state and the District of Columbia over the 2000–2019 sample period. Each row is a state and each column a calendar year. Light-blue cells denote years before an expansion state adopted Medicaid expansion, dark-navy cells denote years from adoption onward, and grey cells denote states that never expanded during the sample period and serve as the comparison group. Expansion states are ordered by adoption year (2014 wave first, then the 2015, 2016, and 2019 adopters), with never-expansion states grouped at the bottom. This staggered rollout is the identifying variation for the event-study design described in Section 5. Expansion dates are from the Kaiser Family Foundation status-of-expansion tracker (KFF 2026).

Figure 2: Total Matched Residency Positions by Expansion Cohort



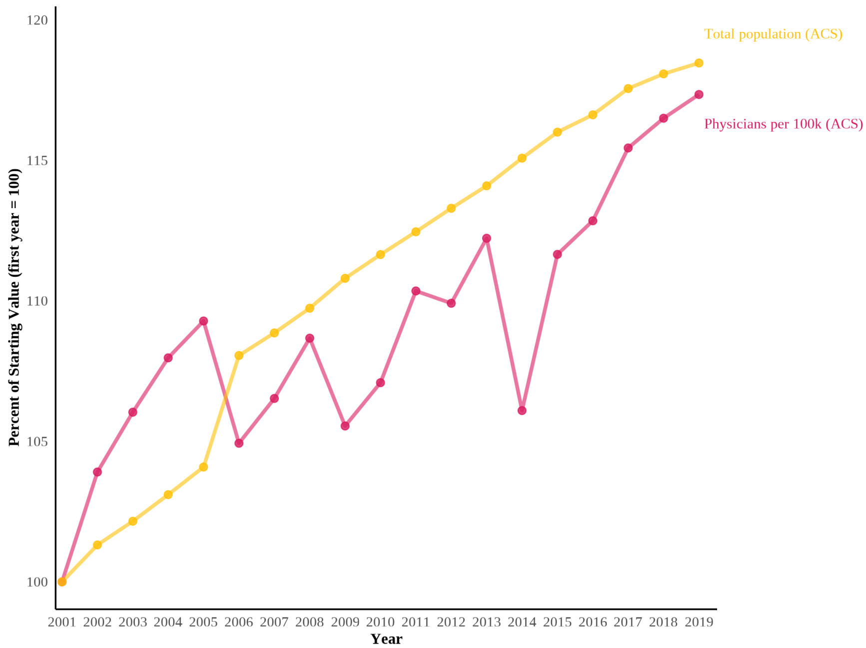
Notes: The figure plots the total number of matched residency positions by year, separately for each Medicaid-expansion cohort. A cohort is defined by the calendar year in which a program's state expanded Medicaid (2014, 2015, 2016, and 2019); programs in states that never expanded over the sample period are grouped as "Never Treated." Each series sums matched positions across all programs in the cohort, where positions are first aggregated to the institution-year level across specialties. Dashed vertical lines mark the expansion years of the treated cohorts. The figure is descriptive and is not adjusted for fixed effects or covariates. Data are from NRMP Main Residency Match reports and newly digitized archival results books, 2000–2019.

Figure 3: Matched Residency Positions per 100,000 Population by Expansion Cohort



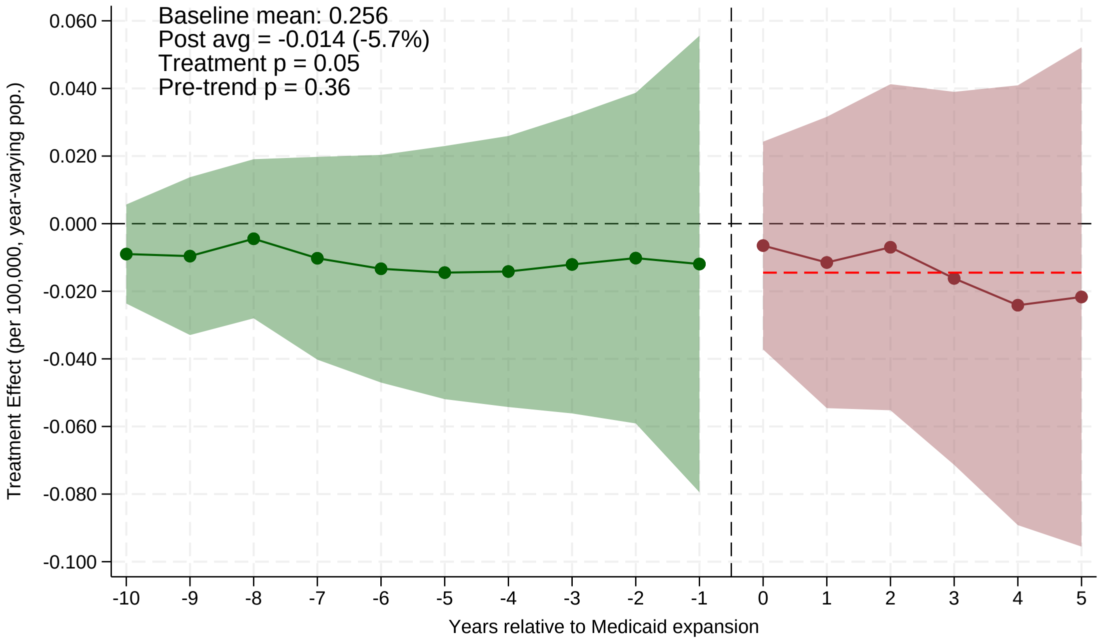
Notes: The figure plots matched residency positions per 100,000 population by year, separately for each Medicaid-expansion cohort. Cohorts are defined by the calendar year in which a program's state expanded Medicaid (2014, 2015, 2016, and 2019), with programs in non-expansion states grouped as "Never Treated." For each cohort and year, matched positions and population are summed across the cohort's states before computing the rate per 100,000, using contemporary (year-varying) state population. Dashed vertical lines mark the expansion years of the treated cohorts. The figure is descriptive and is not adjusted for fixed effects or covariates. Data are from NRMP Main Residency Match reports and newly digitized archival results books, 2000–2019, and U.S. Census Bureau state population estimates.

Figure 4: Growth in Physicians per 100,000 Population Relative to Total Population



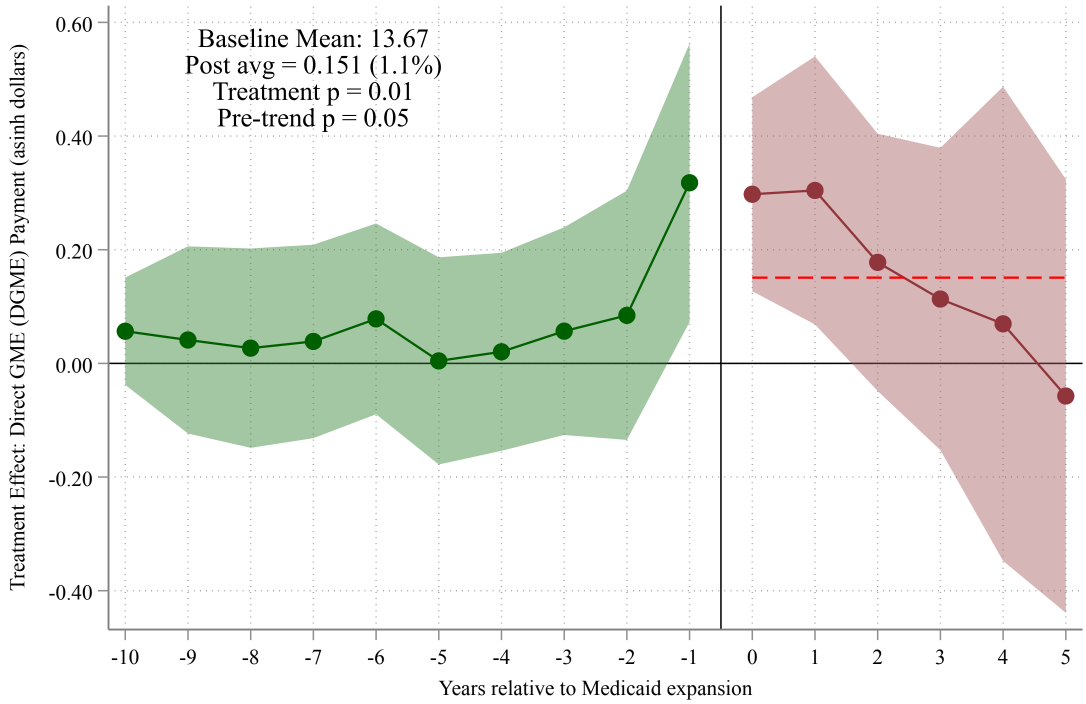
Notes: The figure plots the cumulative percent change since 2001 (2001 = 100) in two national series from the American Community Survey (ACS): physicians per 100,000 population and total population. Physicians are defined as employed workers in the harmonized occupation code for physicians and surgeons (`OCC1950` = 75). Estimates use the ACS person weight (`PERWT`). Data are from IPUMS-USA. The figure motivates the analysis by showing that the physician-to-population ratio has roughly tracked, rather than outpaced, population growth.

Figure 5: Effect of Medicaid Expansion on Matched Residency Positions: Event Study Estimates



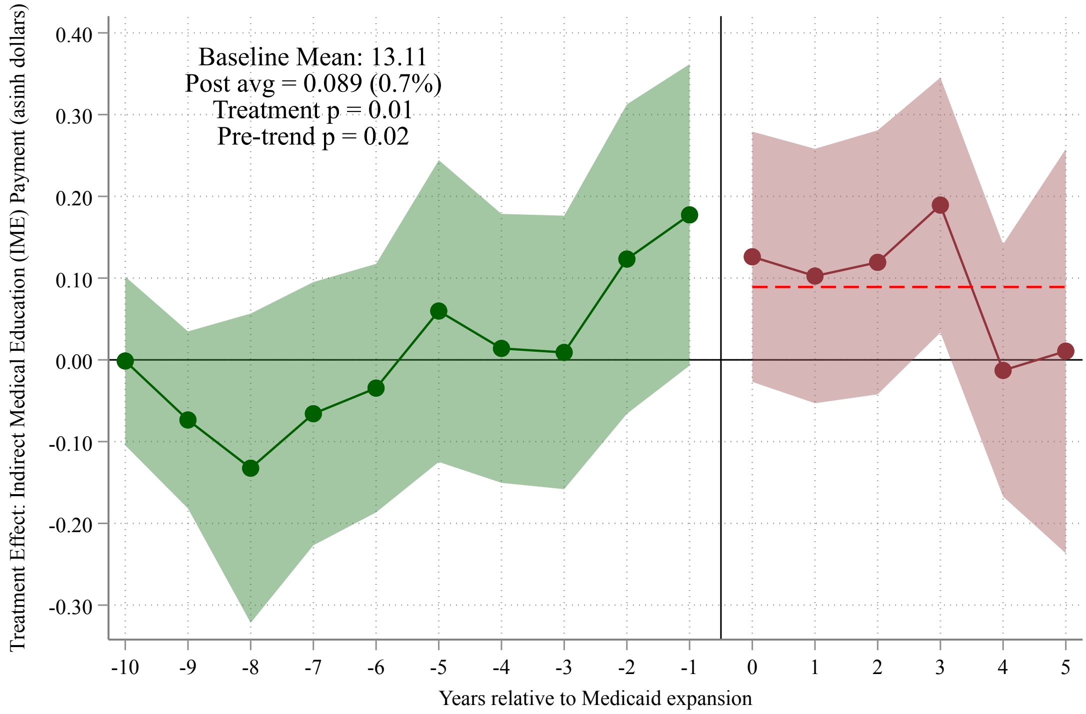
Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), where the outcome is matched residency positions per 100,000 *contemporary* (year-varying) state population from the American Community Survey. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. Green circles (years -10 to -1) show pre-expansion coefficients. Dark red circles (years 0 to $+5$) show post-expansion treatment effects. The dashed red line indicates the average post-expansion treatment effect (-0.014, s.e. 0.026; 95 percent CI $[-0.066, +0.037]$), approximately -5.7 percent relative to the pre-expansion mean of 0.26 matched positions per 100,000 (joint post-coefficients test $p = 0.05$); the ten pre-expansion coefficients are jointly flat (pre-trend $p = 0.36$). The sample is the full 2000–2019 institution panel (1,234 sponsoring institutions) with activity-window coding, so entry and exit do not enter the intake-margin estimate. As discussed in the text, the average effect is statistically indistinguishable from zero; the reallocation decomposition in Section 6.2 is the paper's central result. Shaded bands are 95 percent confidence intervals. Regressions include institution and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

Figure 6: Effect of Medicaid Expansion on Direct GME (DGME) Payments: Event Study Estimates



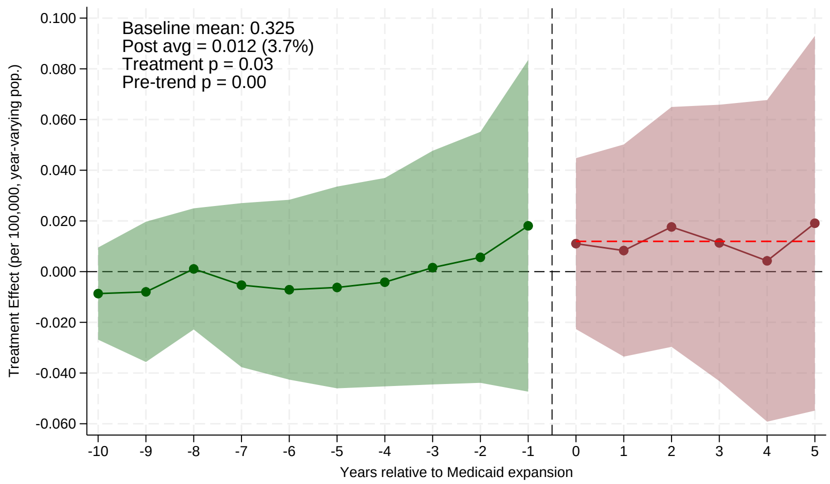
Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024). The outcome is the inverse hyperbolic sine (asinh) of Direct GME (DGME) payments per teaching hospital, from CMS Medicare hospital cost reports; DGME reimburses the direct costs of residency training (resident salaries, benefits, and faculty supervision). Because $\text{asinh}(x) \approx \ln(2x)$ for large x, coefficients approximate proportional changes. The sample comprises all hospitals receiving Medicare GME payments, observed at the hospital–fiscal-year level, with never-expansion states as the comparison group. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. Green circles (years -10 to -1) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion treatment effects. The dashed red line indicates the average post-expansion effect of 0.151, roughly a 15 percent increase in DGME payments in this asinh specification (joint $p < 0.01$; ten-period pre-trend $p = 0.05$); cost-report periods are annualized by months covered. The asinh estimate is not robust to functional form (Chen and Roth 2024): estimated in logs on the subsample with positive payments (98 percent of hospital-years), the payment response disappears (-4.3 percent, $p = 0.57$; Appendix Table A.1); this figure should therefore be read as descriptive of the conventional specification, not as an established revenue channel. Shaded bands are 95 percent confidence intervals. Estimates are based on 1,916 hospitals across 51 states and the District of Columbia. Regressions include hospital and year fixed effects and are unweighted. Standard errors are clustered at the state level.

Figure 7: Effect of Medicaid Expansion on Indirect Medical Education (IME) Payments: Event Study Estimates

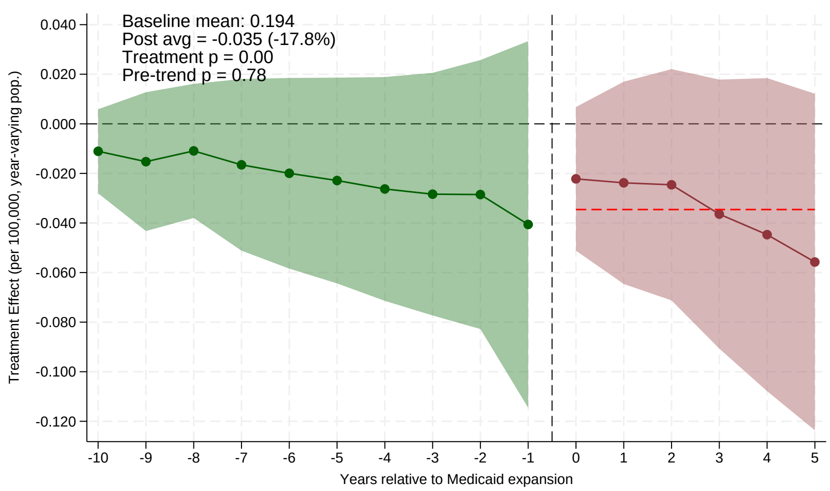


Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024) for the indirect component of GME payments, complementing the Direct GME estimate in Figure 6. The outcome is the inverse hyperbolic sine (asinh) of Indirect Medical Education (IME) payments per teaching hospital, from CMS Medicare hospital cost reports; IME is the indirect operating-cost adjustment paid to teaching hospitals, as distinct from the direct training costs captured by DGME. Because $\operatorname{asinh}(x) \approx \ln(2x)$ for large x, coefficients approximate proportional changes. The average post-expansion effect is 0.089, roughly a 9 percent increase (joint $p < 0.01$), but the ten-period pre-trends test rejects (pre-trend $p = 0.02$); cost-report periods are annualized by months covered. Estimated in logs on the subsample with positive payments, there is no IME response (-5.7 percent, $p = 0.55$; Appendix Table A.1), so the asinh response should not be read as evidence of a revenue channel. The sample comprises all hospitals receiving Medicare GME payments at the hospital–fiscal-year level, with never-expansion states as the comparison group. Green circles (years -10 to -1) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion treatment effects. Shaded bands are 95 percent confidence intervals. Estimates are based on 1,916 hospitals across 51 states and the District of Columbia. Regressions include hospital and year fixed effects and are unweighted. Standard errors are clustered at the state level.

Figure 8: Mechanism: Effect of Medicaid Expansion by State Medicaid GME Formula



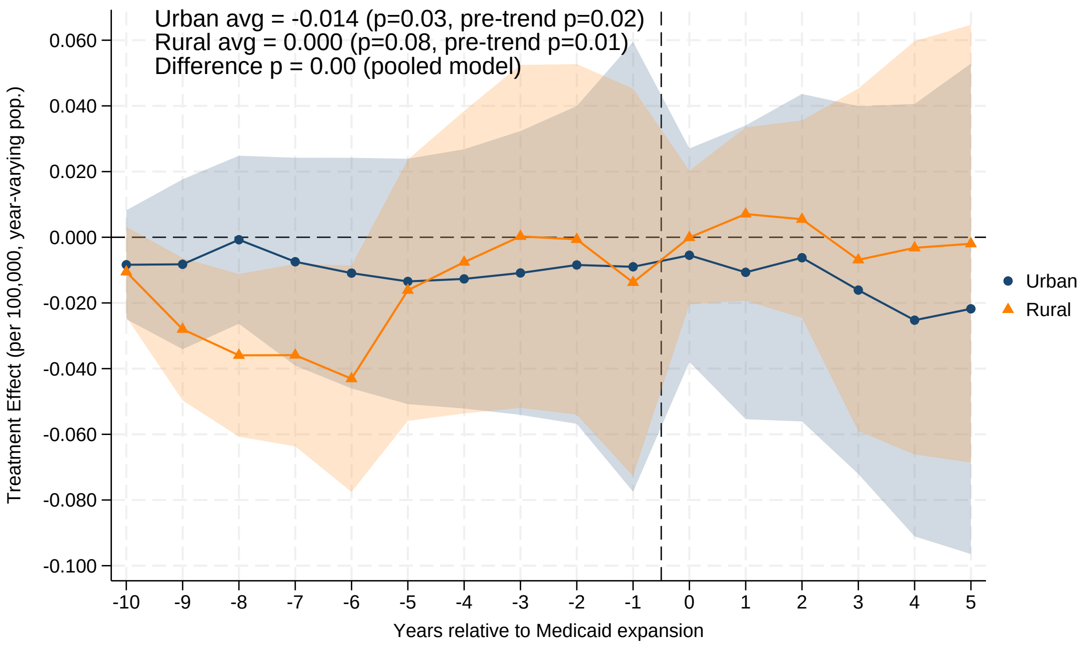
(a) Volume-Responsive GME States



(b) Non-Responsive GME States (Fixed/None)

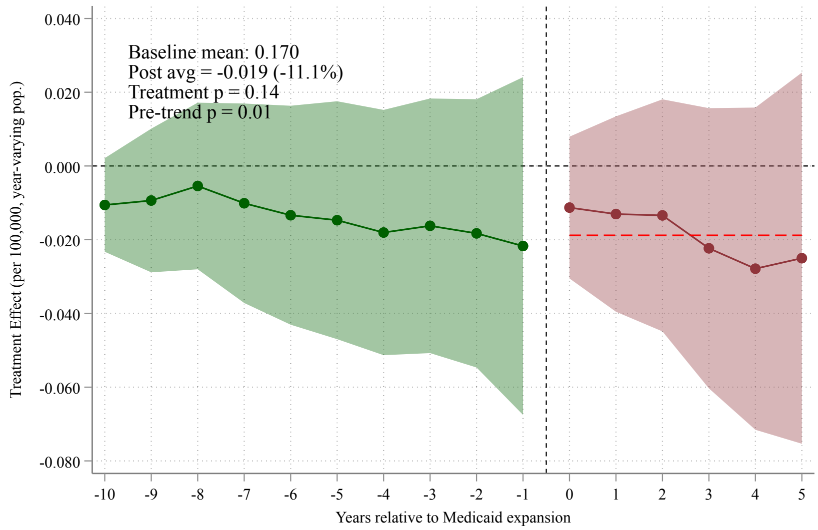
Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), estimated separately for expansion states with volume-responsive (top) versus non-responsive (bottom) Medicaid GME formulas, each using non-expansion states as the comparison group. The outcome is matched residency positions per 100,000 contemporary (year-varying) state population, on the full 2000–2019 panel with activity-window coding. In non-responsive states the average post-expansion effect is -0.035 (a 17.8 percent decline; s.e. 0.025; joint $p < 0.001$) following ten flat pre-expansion coefficients (pre-trend $p = 0.78$); in volume-responsive states it is $+0.012$ ($+3.7$ percent; s.e. 0.026), but that panel *fails* its ten-period pre-trends test ($p < 0.001$), so its level is not interpretable as a treatment effect. A pooled interacted model puts the volume-minus-non-responsive difference at 0.049 per 100,000 (clustered $p < 0.001$; wild cluster bootstrap $p = 0.53$). As Section 6.3 documents, we report this contrast as a negative result. Volume-responsive states reward higher Medicaid volume with larger GME payments, so expansion mechanically raises GME revenue; non-responsive states (fixed per-resident amounts, lump-sum grants, or no Medicaid GME) gain none. States are classified by the Medicaid GME payment rules in force during the treatment window, from the survey of Henderson (2016). Green circles (years -10 to -1) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion treatment effects; the dashed red line marks the average post-expansion effect. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects, are weighted by state population. Standard errors are clustered at the state level.

Figure 9: Effect of Medicaid Expansion on Matched Residency Positions by Hospital Location: Event Study Estimates

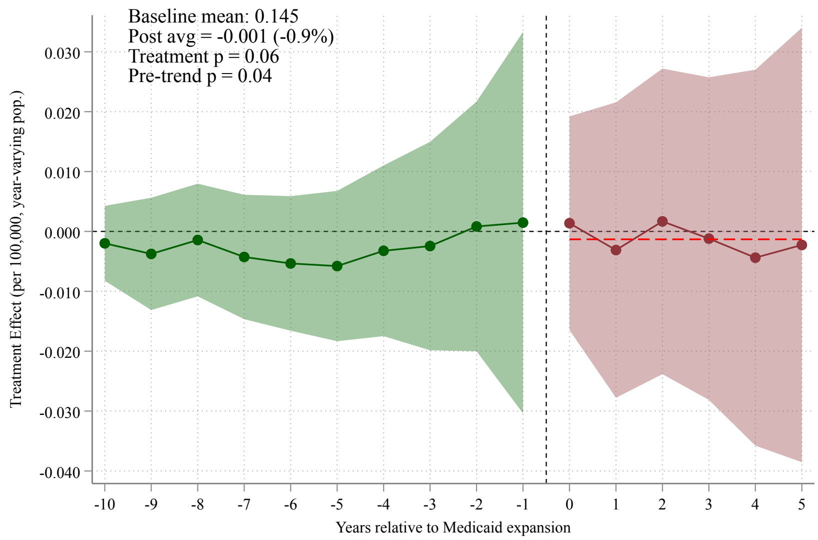


Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), estimated in two split-sample regressions for urban (blue circles) and rural (orange triangles) hospitals; each subsample keeps the never-expansion hospitals of its own type as the comparison group, so each series carries its own pre-expansion path and pre-trends test. Urban hospitals are those whose institution ZIP code (obtained by geocoding the institution address) carries a 2010 RUCA code of 1–3; rural hospitals are those with ZIP-level RUCA codes above 3. RUCA is unavailable for institutions that exited before 2010 and for Alaska and Hawaii institutions (excluded from this split only). The outcome is matched positions per 100,000 contemporary (year-varying) population. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. The average post-expansion treatment effect is -0.014 for urban hospitals (-5.0 percent; s.e. 0.027; joint $p = 0.03$; ten-period pre-trend $p = 0.02$) and $+0.0001$—zero to three decimal places—for rural hospitals (joint $p = 0.08$; pre-trend $p = 0.009$). Both subsamples fail their ten-period pre-trends tests, so the split is read descriptively. A pooled interacted model puts the urban-minus-rural difference in average post effects at -0.045 ($p < 0.001$), but that model shares pre-trends across groups. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

Figure 10: Effect of Medicaid Expansion on Matched Residency Positions by Specialty Type: Event Study Estimates



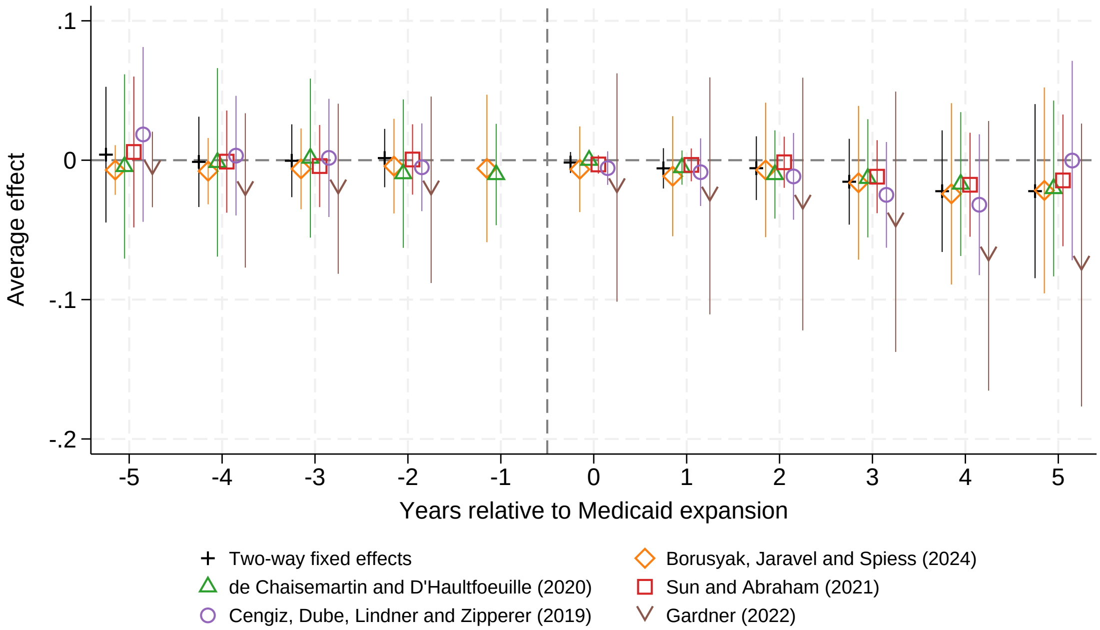
(a) Non-Primary Care Positions



(b) Primary Care Positions

Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024) estimated separately for non-primary care (top panel) and primary care specialties (bottom panel). Primary care includes family medicine, internal medicine, and pediatrics. The outcome is matched positions per 100,000 contemporary (year-varying) population. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. Estimation is at the institution-by-specialty-group level, each group carrying its own unit fixed effects. For non-primary care programs, the average post-expansion treatment effect is -0.019 (-11.1 percent; s.e. 0.017; joint $p = 0.14$; pre-trend $p = 0.01$). For primary care programs, it is -0.001 (-0.9 percent; s.e. 0.014; joint $p = 0.06$; pre-trend $p = 0.04$). Both panels exhibit differential ten-period pre-trends, so the specialty estimates should be read as suggestive; primary-care intake at existing institutions shows no contraction. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

Figure 11: Robustness of Event Study Estimates Across Difference-in-Differences Estimators: Matched Residency Positions per 100,000 Population



Notes: The figure compares event-study estimates of the effect of Medicaid expansion on matched residency positions per 100,000 contemporary (year-varying) population across six difference-in-differences estimators that are robust to treatment-effect heterogeneity under staggered adoption: two-way fixed effects; Borusyak, Jaravel, and Spiess (2024); de Chaisemartin and D'Haultfoeuille (2020); Sun and Abraham (2021); Cengiz, Dube, Lindner, and Zipperer (2019); and Gardner (2022). The heterogeneity-robust estimator of Callaway and Sant'Anna (2021) is omitted because it does not accept the population weights the other estimators use and its never-treated comparison cells are sparse in this sample; estimated unweighted with the not-yet-treated control group it gives a simple ATT of -0.005 per 100,000 (s.e. 0.007, $p = 0.46$), consistent with the estimates shown. The horizontal axis measures years relative to a state's Medicaid expansion; the dashed vertical line marks the expansion year. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

ONLINE APPENDIX

Does Coverage Expansion Grow the Physician Pipeline? Evidence from the ACA

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A Tables

Table A.1: Payment Responses Decomposed: Intensive and Extensive Margins

	Coefficient	(s.e.)	p-value
Panel A: Direct GME payments			
Log payments (positive subsample), post-expansion effect	−0.043	(0.077)	0.57
Log payments (positive subsample), volume-arm interaction	−0.106	(0.072)	0.14
Any payment (extensive margin), post-expansion effect	−0.009	(0.005)	0.11
Any payment (extensive margin), volume-arm interaction	0.000	(0.004)	0.94
Panel B: Indirect medical education payments			
Log payments (positive subsample), post-expansion effect	−0.057	(0.093)	0.55
Log payments (positive subsample), volume-arm interaction	−0.109	(0.106)	0.30
Any payment (extensive margin), post-expansion effect	−0.001	(0.003)	0.79
Any payment (extensive margin), volume-arm interaction	0.001	(0.003)	0.80

Notes: The table decomposes the payment responses that the inverse-hyperbolic-sine specifications measure elsewhere in the paper into an intensive and an extensive margin; asinh point estimates are sensitive to functional-form choices that this decomposition avoids (Chen and Roth 2024). The log rows estimate the natural log of annualized Medicare cost-report payments on the subsample of hospital–fiscal-years with positive payments (98 percent of the estimation sample for Direct GME, 89 percent for IME), so coefficients are proportional changes on the intensive margin. The extensive-margin rows estimate a linear probability model for whether a hospital receives any payment, so coefficients are changes in percentage-point terms. All rows use a static treated-post indicator with hospital and fiscal-year fixed effects, unweighted; the volume-arm interaction rows report the coefficient on treated-post interacted with the volume-responsive formula classification—the cross-arm difference. Standard errors are clustered at the state level.

Table A.2: The Cross-Arm Difference: Estimates, Sensitivity, and Inference

	Estimate	(s.e.)	p-value
Panel A: Event-study estimate and classification sensitivity			
Baseline (2015 classification)	0.049	(0.014)	< 0.001
Flipping Maryland	0.058	(0.014)	< 0.001
Flipping Minnesota	0.050	(0.014)	< 0.001
Flipping Montana	0.049	(0.014)	< 0.001
Flipping Iowa	0.051	(0.014)	< 0.001
Flipping New Mexico	0.051	(0.014)	< 0.001
Panel B: Alternative design			
Timing-only design (not-yet-treated controls)	0.045	(0.013)	< 0.001
Panel C: Inference on the baseline difference			
Static TWFE analog, clustered	0.050	(0.029)	0.08
Static TWFE analog, wild cluster bootstrap-t	0.050		0.53
Randomization inference (500 draws)	0.047		0.48

Notes: The table collects the volume-minus-non-responsive difference in average post-expansion effects on matched positions per 100,000 contemporary population—the mechanism section's central object (Section 6.3)—across specifications and inference standards. Panel A reports the baseline pooled interacted event-study estimate (imputation estimator with institution and year fixed effects, 2010-population weights, state-clustered standard errors, delta-method p-values) and its value when each of the five states whose formula coding required judgment is flipped to the opposite arm, one at a time. Panel B re-estimates the difference inside the timing-only design that uses only eventually-treated states (not-yet-treated controls). Panel C reports the static two-way-fixed-effects analog of the difference under clustered inference and under wild cluster bootstrap-t (Webb weights, 9,999 replications), and the event-study difference under randomization inference permuting expansion-timing cohorts across states (500 draws; Section 7.5). The difference survives every classification perturbation but is indistinguishable from zero under bootstrap and randomization inference.

Table A.3: Baseline Balance Across Medicaid GME Formula Groups (2010)

Baseline characteristic (2010)	Volume-responsive (1)	Fixed/none (2)	Never-expansion (3)	Difference (1)−(2)	p-value
Matched positions per program	35.9	30.0	37.0	5.9	0.332
Offered positions (quota) per program	38.3	31.4	38.8	6.9	0.264
Share of programs filling every offered position	0.83	0.75	0.87	0.07	0.451
Matched positions per 100,000 population	10.10	7.72	5.46	2.38	0.437
Program-level mean, population-weighted	0.35	0.20	0.37	0.15	–
Offered positions (quota) per 100,000	10.66	8.03	5.76	2.63	0.402
Number of programs	14.1	16.2	8.9	-2.1	0.748
State population (millions)	5.5	6.6	6.3	-1.2	0.697
Rural program share	0.05	0.08	0.08	-0.03	0.519
Primary-care share of matched positions	0.52	0.58	0.57	-0.06	0.300
Mean expansion year	2014.6	2014.3	–	0.3	0.416
Number of states	21	13	17		

Notes: The table reports baseline (2010, pre-treatment) means by Medicaid GME formula group, for the same characteristics as the summary-statistics table (Table 1). Expansion states are classified as volume-responsive or fixed/none by the payment rules in force during the 2014–2019 treatment window, from the survey of Henderson (2016); never-expansion states are shown for reference. All rows except the population-weighted row are unweighted means across states (the state is the unit of observation); the p-value is from a Welch two-sample t-test of the volume-responsive versus fixed/none difference at the state level, so power is limited by the number of states. The population-weighted row reports the program-level mean of matched positions per 100,000 (2010 population), weighted by state population—the object the estimation weights by; its cross-group gap reflects the concentration of training in a few large volume-responsive states rather than state-level imbalance. The fill-share row is measured at the NRMP specialty-program level (the share of a state's specialty programs filling every offered position); the primary-care share is the share of matched positions in family medicine, internal medicine, and pediatrics; rural programs are those whose institution ZIP code (obtained by geocoding the institution address) carries a 2010 RUCA code above 3; RUCA is unavailable for institutions that exited before 2010 and for Alaska and Hawaii institutions, whose coordinates fall outside the geocoder's continental-U.S. filter, so those institutions are excluded from the rural/urban split (not from the main analysis). Data are from NRMP Main Residency Match reports and the U.S. Census Bureau.

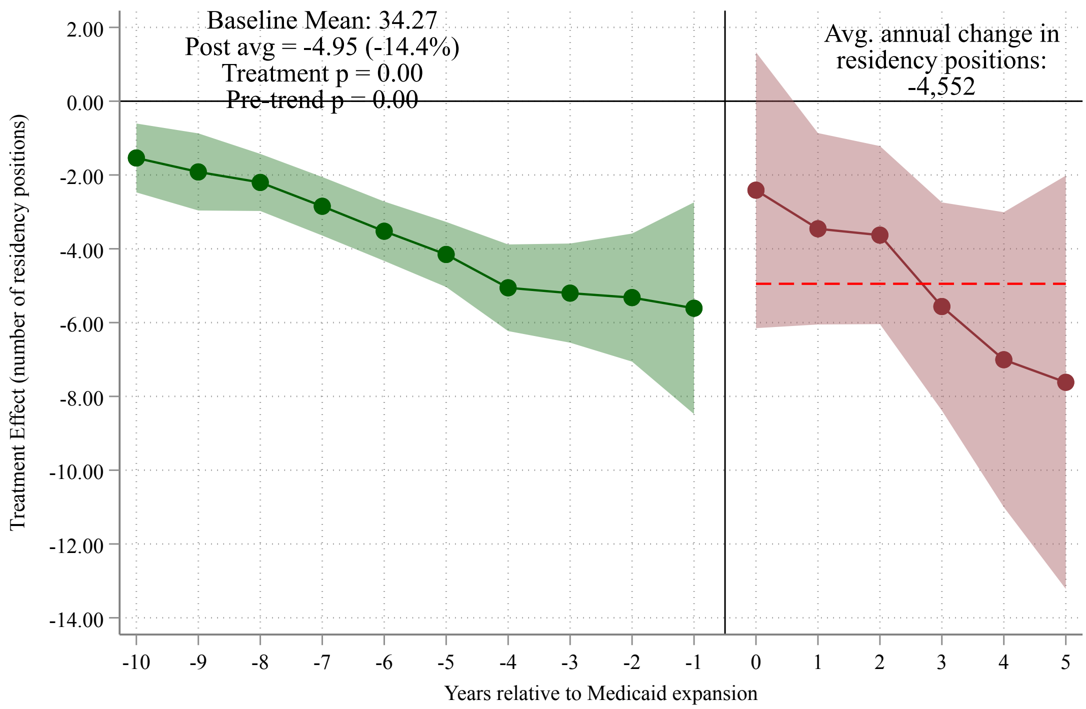
Table A.4: Multiple-Testing Adjustment for the Primary Family of Per-Capita Effects

	Clustered		Randomization inference	
Hypothesis (avg. post effect, matched per 100,000)	p-value	q-value	p-value	q-value
Full sample (headline)	0.052	0.094	0.911	0.996
Urban hospitals	0.025	0.076	0.918	0.996
Rural hospitals	0.078	0.100	0.996	0.996
Primary care	0.063	0.094	0.986	0.996
Non-primary care	0.141	0.158	0.573	0.996
Offered positions (quota)	0.209	0.209	0.940	0.996
Volume-responsive arm	0.035	0.078	0.694	0.996
Non-responsive arm	<0.001	<0.001	0.737	0.996
Cross-formula difference	<0.001	0.003	0.479	0.996

Notes: The table reports Benjamini–Hochberg false-discovery-rate q-values for the primary family of average post-expansion effects on matched positions per 100,000 population, computed under two inference standards. "Clustered" p-values are the joint tests of the post-expansion event-time coefficients with standard errors clustered at the state level (the offered-positions row uses the quota outcome). "Randomization inference" p-values permute the vector of state expansion-timing cohorts across the states in each specification's sample under the sharp null of no effect (1,000 draws for the full-sample headline, 500 for the other members) and compare the observed average post effect with the permutation distribution; studentized variants that permute the t-statistic instead of the coefficient are reported in the replication output (`ri-yearvarying-summary.csv`, `ri-extended-summary.csv`) and give the same conclusions. Within each standard, q-values apply the Benjamini–Hochberg correction across the family of nine tests (the two mechanism arms and the cross-arm difference were added at referee request); a q-value below 0.05 indicates a finding that survives a 5 percent false-discovery-rate correction. No member survives under randomization inference (all

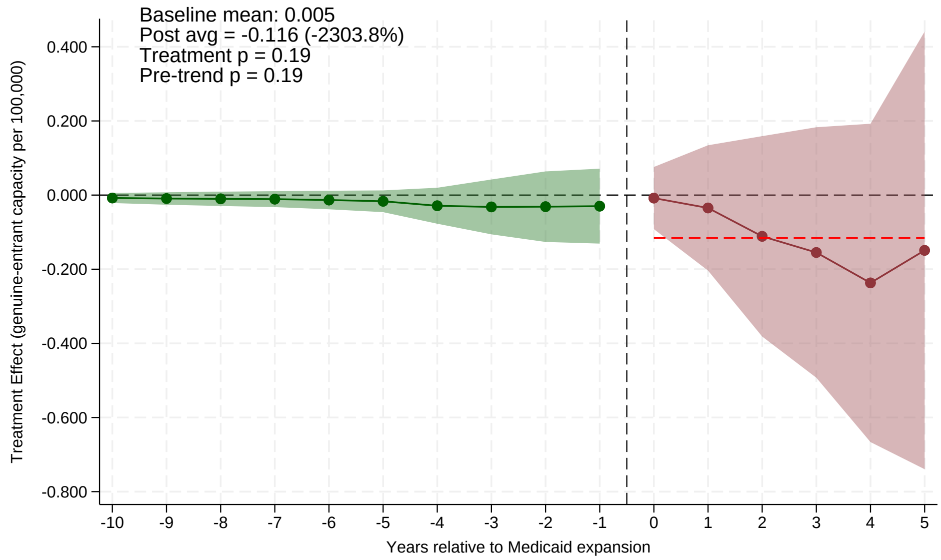
B Figures

Figure A.1: Effect of Medicaid Expansion on Matched Residency Positions in Levels: Event Study Estimates



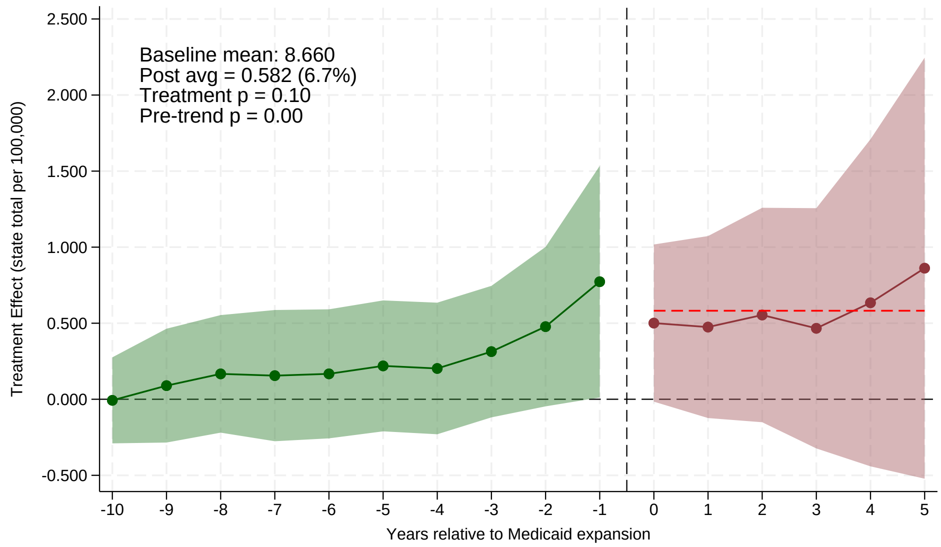
Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), where the outcome is the absolute number of matched residency positions per hospital (not normalized by population). The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. Green circles (years -10 to -1) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion treatment effects. The dashed red line indicates the average post-expansion treatment effect of -3.85 positions per hospital, corresponding to a 13.1 percent decline relative to the pre-expansion mean of 29.43 matched positions ($p < 0.01$). **The pre-expansion coefficients in this raw-count specification are large and negative** (on the order of -2 positions per hospital, comparable in magnitude to the treatment effect), indicating a violation of parallel trends. This levels estimate therefore corroborates the *direction* of the effect but should not be read as a clean causal magnitude; the paper's primary specification instead normalizes by contemporary population, which restores flat pre-trends (see Section 7.5). Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.2: Genuine-Entrant Capacity: Event Study



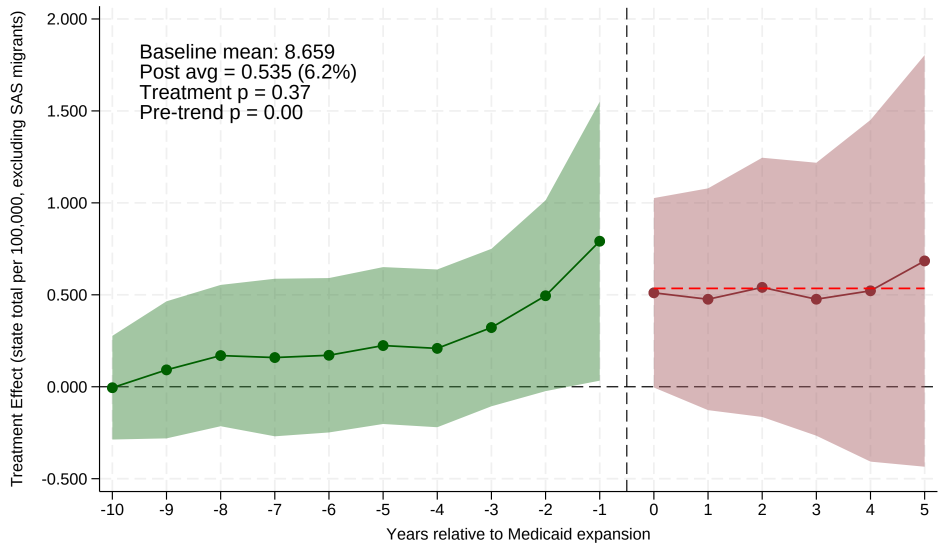
Notes: The figure estimates the state-level event study for matched positions per 100,000 contemporary population at genuinely new post-2010 entrant institutions only—the direct test of whether Medicaid expansion induced new sponsors to enter training provision (Section 6.2). Because pre-expansion capacity at post-2010 entrants is near zero by construction, effects are read in levels; the annotation box reports the average post-expansion effect, joint post-period p-value, and ten-period pre-trends p-value, and the unweighted counterpart appears in Table 2. Estimates use the imputation estimator of Borusyak, Jaravel, and Spiess (2024) with state and year fixed effects, 2010-population weights, and state-clustered standard errors; entry estimates are reported under clustered inference only (Section 7.5). Green circles show pre-expansion coefficients; dark red circles show post-expansion effects; shaded bands are 95 percent confidence intervals. Data: NRMP Main Residency Match reports; ACGME Accreditation Data System.

Figure A.3: State-Level Training Totals: Event Study Including the Entry Margin



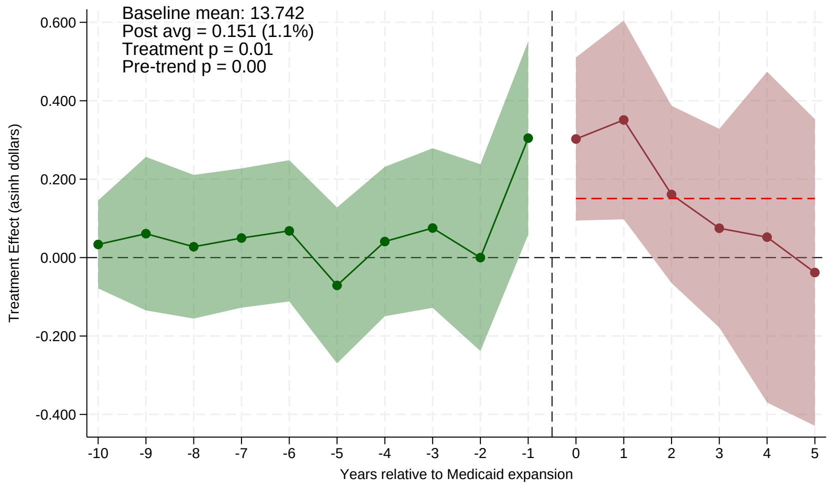
Notes: The figure aggregates matched positions to state-level totals per 100,000 contemporary population—the object that correctly includes the entry margin, since an entrant institution's pre-entry contribution to its state's total is genuinely zero—and re-estimates the headline event study with the state as the unit (51 states). The average post-expansion effect is +0.58 per 100,000 (s.e. 0.42; +6.7 percent of the pre-expansion mean; joint $p = 0.10$); restricting the totals to always-active institutions gives +11.3 percent (joint $p = 0.34$). The state-level pre-trend test fails, so the magnitude should be read with caution; the sign contrast with the intake-margin estimates is the robust object (Section 6.2). Estimates use the imputation estimator of Borusyak, Jaravel, and Spiess (2024) with state and year fixed effects, 2010-population weights, and state-clustered standard errors. Green circles show pre-expansion coefficients; dark red circles show post-expansion effects; shaded bands are 95 percent confidence intervals.

Figure A.4: State Totals Excluding Single-Accreditation Migrants

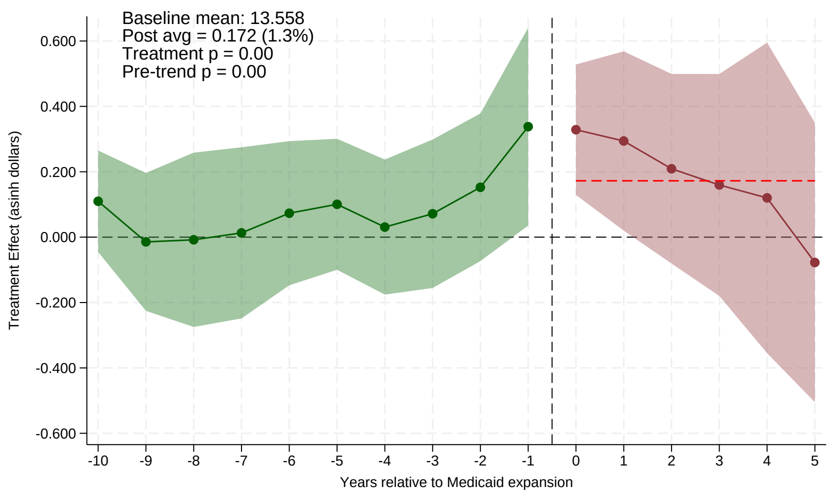


Notes: The figure re-estimates the state-level total event study of Figure A.3 after removing all matched positions at the 100 post-2010 entrant institutions classified as AOA-to-ACGME Single Accreditation System migrants (Section 4), so the totals reflect existing institutions plus genuinely new entrants only. The annotation box reports the average post-expansion effect, its percent-of-baseline equivalent, the joint post-period p-value, and the ten-period pre-trends p-value; the corresponding unweighted estimate appears in Table 2. The non-declining state total is not an artifact of the accreditation transition. Estimates use the imputation estimator of Borusyak, Jaravel, and Spiess (2024) with state and year fixed effects, 2010-population weights, and state-clustered standard errors. Green circles show pre-expansion coefficients; dark red circles show post-expansion effects; shaded bands are 95 percent confidence intervals. Data: NRMP Main Residency Match reports; ACGME Accreditation Data System.

Figure A.5: Group-Specific Payment Evidence: Effect of Medicaid Expansion on Direct GME (DGME) Payments by State Medicaid GME Formula



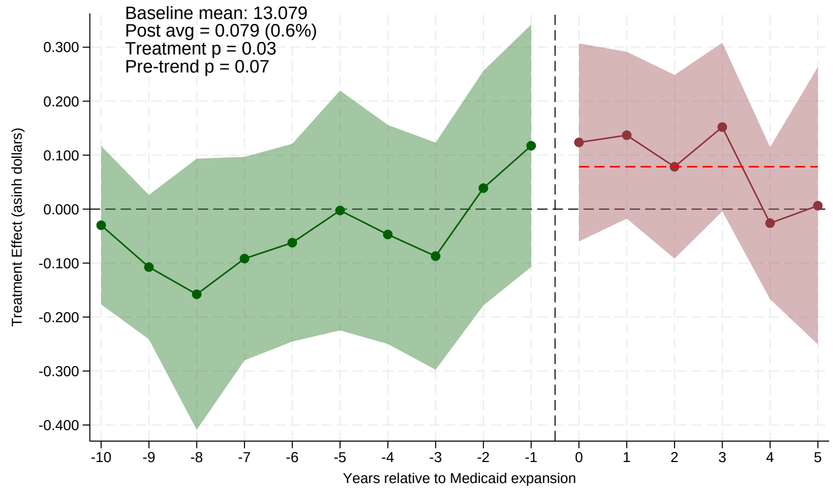
(a) Volume-Responsive GME States



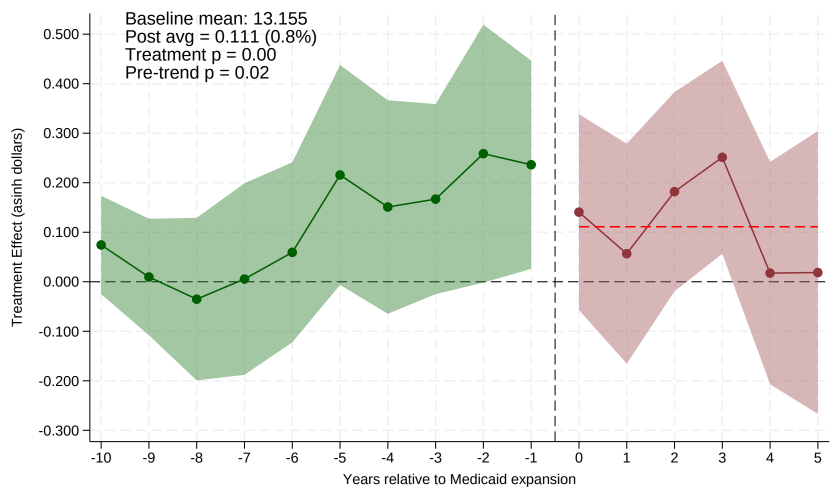
(b) Non-Responsive GME States (Fixed/None)

Notes: The figure re-estimates the Direct GME (DGME) payment event study from Figure 6 separately for volume-responsive (top) and non-responsive (bottom) expansion states, each using never-expansion states as the comparison group. The outcome is Medicare cost-report payments, which we read as a proxy for training-related revenue rather than a direct measure of state Medicaid GME payments (see Section 6.3). The outcome is the inverse hyperbolic sine (asinh) of DGME payments per teaching hospital, so coefficients approximate proportional changes. With cost-report periods annualized by months covered, in volume-responsive states the average post-expansion effect is 0.15 (s.e. 0.13; joint $p < 0.01$) and in non-responsive states it is 0.17 (s.e. 0.15; joint $p < 0.01$); both panels fail their ten-period pre-trend tests (volume-responsive $p < 0.001$, non-responsive $p = 0.004$), and the cross-group difference is null (-0.02, s.e. 0.11, $p = 0.87$ from a pooled interacted model). Estimated in logs on the subsample with positive payments, the cross-arm payment difference is likewise null (Table A.1); see Figures A.7 and A.8 for the FTE/rate decomposition. Green circles (years -10 to -1) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion effects; the dashed red line marks the average post-expansion effect. Shaded bands are 95 percent confidence intervals. Estimates come from CMS Medicare hospital cost reports at the hospital–fiscal-year level. Regressions include hospital and year fixed effects and are unweighted. Standard errors are clustered at the state level.

Figure A.6: Group-Specific Payment Evidence: Effect of Medicaid Expansion on Indirect Medical Education (IME) Payments by State Medicaid GME Formula



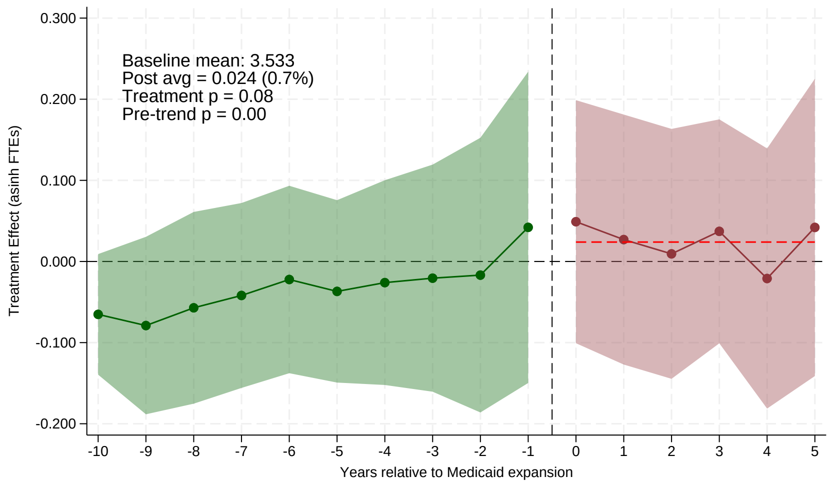
(a) Volume-Responsive GME States



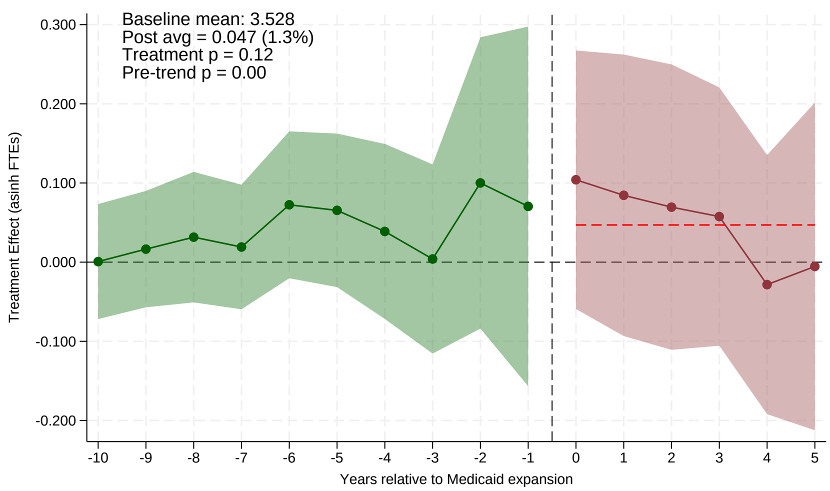
(b) Non-Responsive GME States (Fixed/None)

Notes: The figure re-estimates the Indirect Medical Education (IME) payment event study from Figure 7 separately for volume-responsive (top) and non-responsive (bottom) expansion states, each using never-expansion states as the comparison group. The outcome is the inverse hyperbolic sine (asinh) of IME payments per teaching hospital, from Medicare cost reports (a proxy for training-related revenue; see Section 6.3). With cost-report periods annualized by months covered, in volume-responsive states the average post-expansion effect is 0.08 (s.e. 0.08; joint $p = 0.03$; pre-trend $p = 0.07$) and in non-responsive states it is 0.11 (s.e. 0.09; joint $p < 0.001$; pre-trend $p = 0.02$); the cross-group difference is null (-0.04, s.e. 0.07, $p = 0.56$ from a pooled interacted model). With the log-level null (Table A.1) there is no credible payment gradient. Green circles (years -10 to -1) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion effects. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and are unweighted. Standard errors are clustered at the state level.

Figure A.7: Decomposition of the DGME Payment Response: Resident FTEs by State Medicaid GME Formula



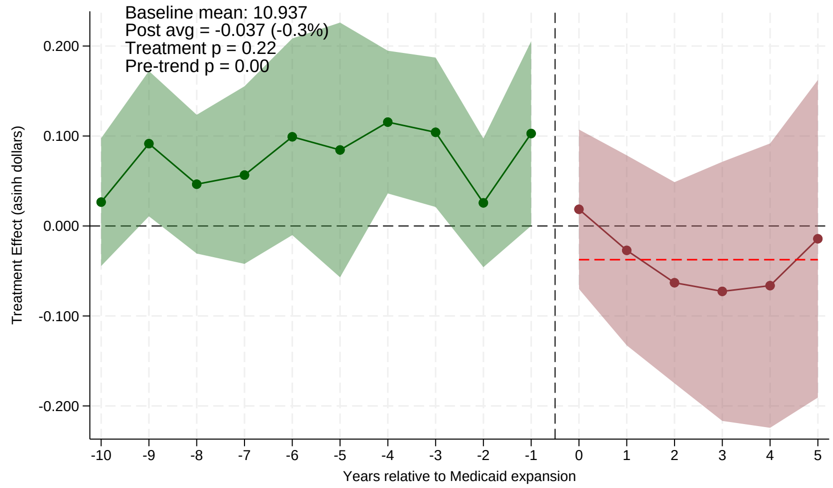
(a) Volume-Responsive GME States



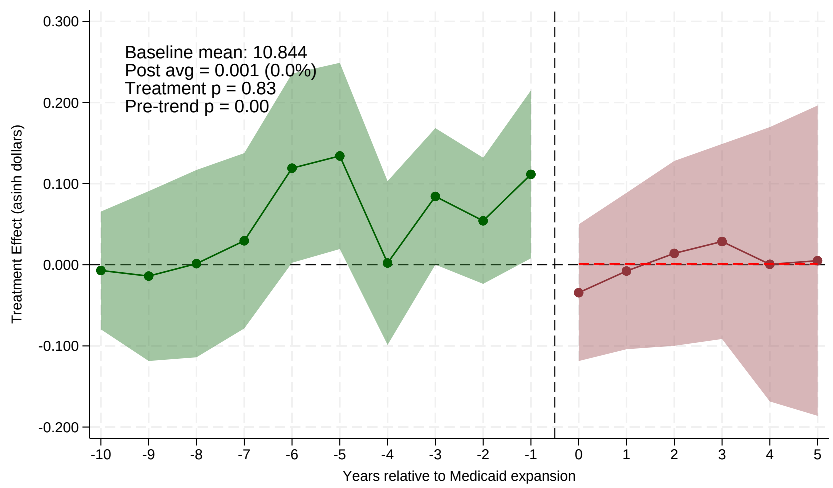
(b) Non-Responsive GME States (Fixed/None)

Notes: The figure decomposes the Direct GME payment response in Figure A.5: the outcome is the inverse hyperbolic sine of cost-report DGME resident FTEs per teaching hospital, so coefficients approximate proportional changes in the number of residents hospitals report for payment purposes. With cost-report periods annualized by months covered, resident FTEs move in *neither* arm: the volume-responsive average post effect is 0.02 (s.e. 0.08; joint $p = 0.08$; pre-trend $p = 0.001$) and the non-responsive effect is 0.05 (s.e. 0.09; joint $p = 0.12$; pre-trend $p = 0.004$); the cross-group difference is null (-0.01, s.e. 0.07, $p = 0.86$). On the linked NRMP–cost-report sample the FTE response is likewise absent (Section 7.4). Green circles (years -10 to -1) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion effects. Shaded bands are 95 percent confidence intervals. Estimates come from CMS Medicare hospital cost reports at the hospital–fiscal-year level with never-expansion states as the comparison group. Regressions include hospital and year fixed effects and are unweighted. Standard errors are clustered at the state level.

Figure A.8: Decomposition of the DGME Payment Response: Payment per Resident FTE by State Medicaid GME Formula



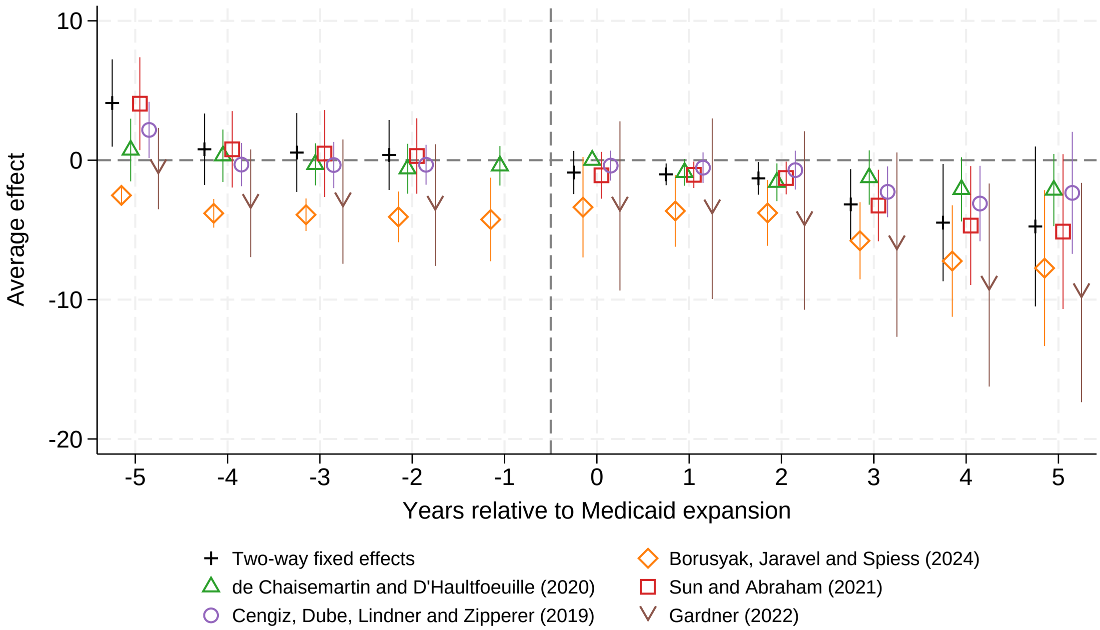
(a) Volume-Responsive GME States



(b) Non-Responsive GME States (Fixed/None)

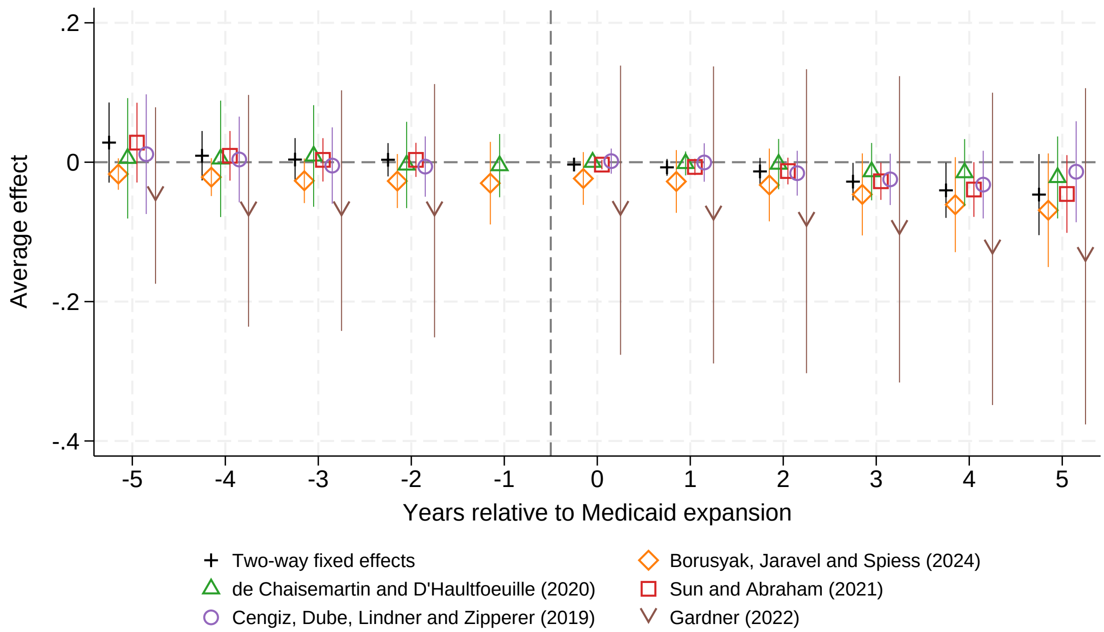
Notes: The figure completes the decomposition in Figure A.7: the outcome is the inverse hyperbolic sine of DGME payment per resident FTE (hospitals with positive FTEs only). Per-FTE payments do not rise in either group: the average post effect is -0.04 in volume-responsive states (s.e. 0.06; joint $p = 0.22$) and 0.00 in non-responsive states (s.e. 0.05; joint $p = 0.83$; pre-trend $p = 0.004$); the cross-group difference is null (-0.03, s.e. 0.04, $p = 0.43$). The volume-responsive panel exhibits a differential pre-trend (pre-trend $p < 0.001$), so its level should be read with caution; with the annualized FTE panel showing no quantity response either (Figure A.7), the decomposition does not support a payment mechanism in any direction. Green circles (years -10 to -1) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion effects. Shaded bands are 95 percent confidence intervals. Estimates come from CMS Medicare hospital cost reports at the hospital–fiscal-year level with never-expansion states as the comparison group. Regressions include hospital and year fixed effects and are unweighted. Standard errors are clustered at the state level.

Figure A.9: Robustness of Event Study Estimates Across Difference-in-Differences Estimators, Total Matched Residency Positions in Levels



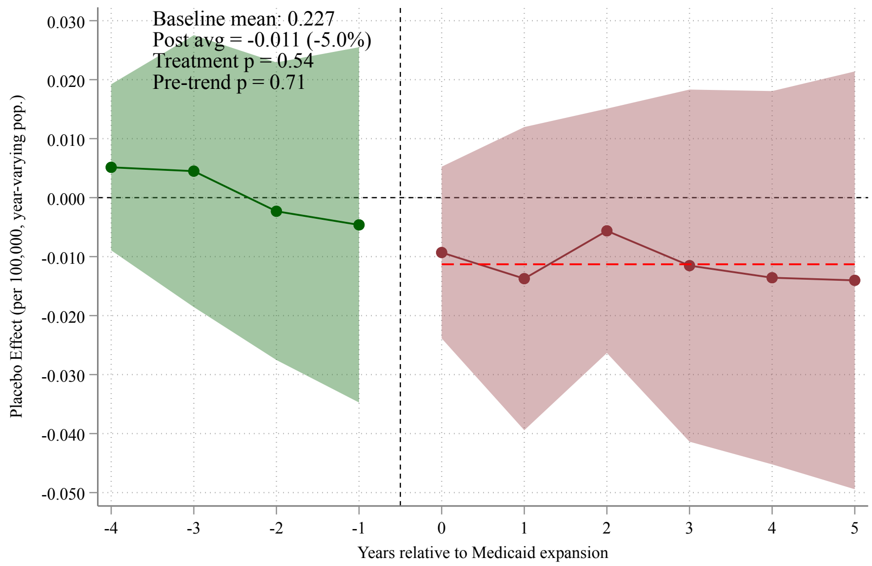
Notes: The figure compares event-study estimates of the effect of Medicaid expansion on the absolute number of matched residency positions per hospital across six difference-in-differences estimators that are robust to treatment-effect heterogeneity under staggered adoption: two-way fixed effects; Borusyak, Jaravel, and Spiess (2024); de Chaisemartin and D'Haultfoeuille (2020); Sun and Abraham (2021); Cengiz, Dube, Lindner, and Zipperer (2019); and Gardner (2022). The heterogeneity-robust estimator of Callaway and Sant'Anna (2021) is omitted because it does not accept the population weights used elsewhere in the paper and its never-treated comparison cells are sparse in this sample; the unweighted not-yet-treated estimate is reported in Section 7.1. The dashed vertical line marks the expansion year. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.10: Robustness of Event Study Estimates Across Difference-in-Differences Estimators, Residency Quota Positions per 100,000 Population

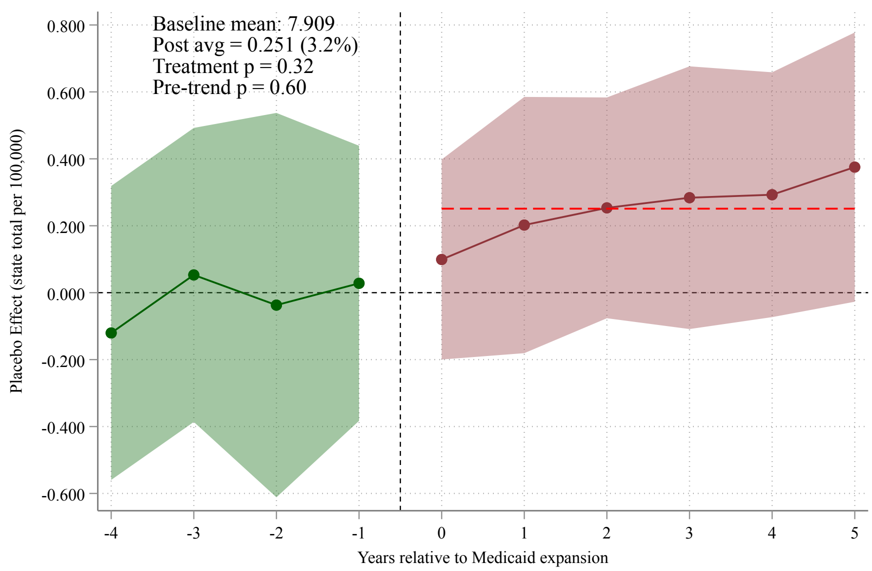


Notes: The figure compares event-study estimates of the effect of Medicaid expansion on quota positions per 100,000 state population across six difference-in-differences estimators. Quota positions measure positions offered by residency programs prior to the matching algorithm. The estimators are: two-way fixed effects; Borusyak, Jaravel, and Spiess (2024); de Chaisemartin and D'Haultfoeuille (2020); Sun and Abraham (2021); Cengiz, Dube, Lindner, and Zipperer (2019); and Gardner (2022); the Callaway and Sant'Anna (2021) estimator is omitted as in the previous figure (population weights and sparse never-treated cells). The dashed vertical line marks the expansion year. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.11: Pre-ACA Placebo: Expansion Dates Shifted Back Ten Years, 2000–2013 Sample



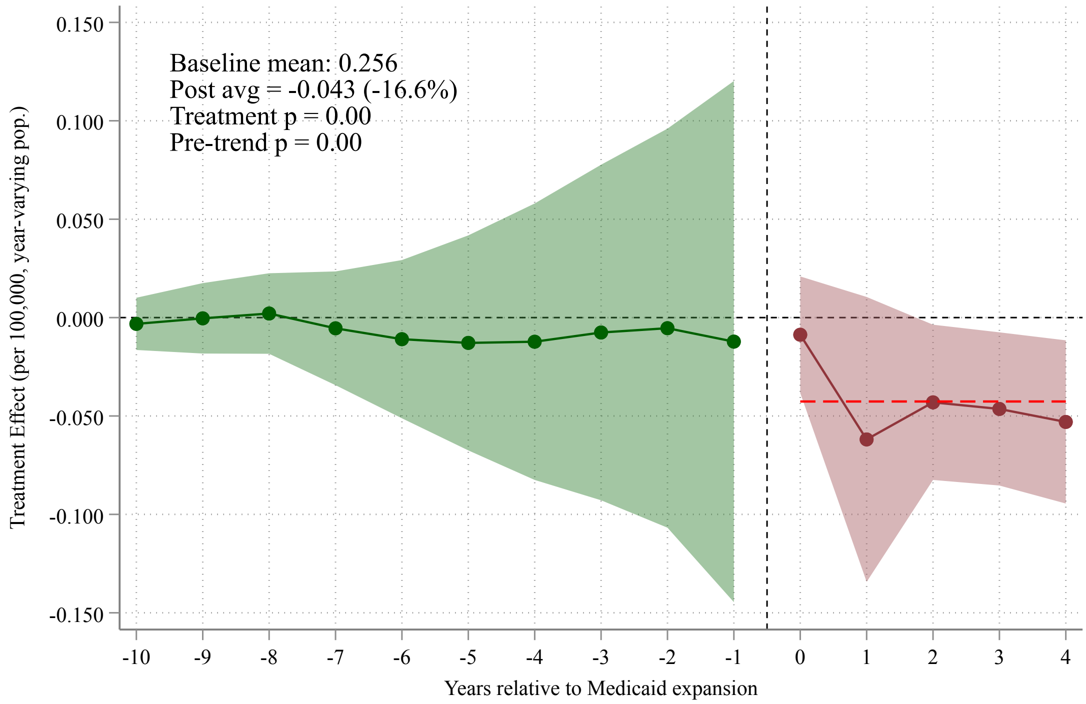
(a) Institution level (activity-window coding)



(b) State-level totals (entry margin included)

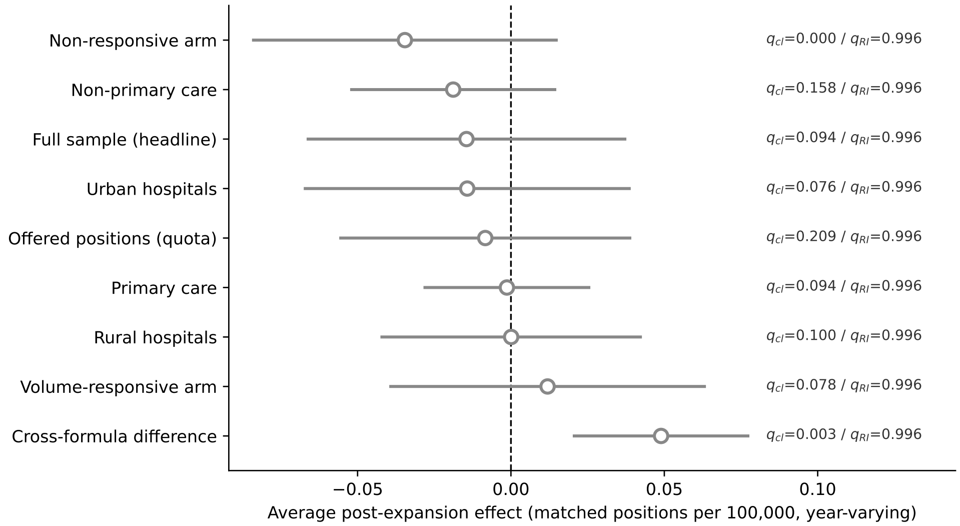
Notes: The figure assigns each expansion state a placebo expansion ten years before its actual one (a 2014 adopter receives a placebo date of 2004) and re-estimates the event study on 2000–2013 data only, so no real treatment enters the sample; never-expansion states serve as controls. If differential regional trends in training capacity—rather than expansion itself—drove the post-2014 estimates, the same pattern should appear around the placebo dates. It does not: at the institution level (panel a) the average placebo effect is -0.011 per 100,000 (s.e. 0.013; joint $p = 0.54$) after flat placebo pre-trends ($p = 0.71$), and at the state level (panel b) it is $+0.25$ (s.e. 0.18; joint $p = 0.32$; pre-trend $p = 0.60$). Sample, outcome construction, weights, estimator, and clustering are identical to the headline design (Figure 5), with four placebo pre-periods and up to six placebo post-periods (later cohorts contribute fewer horizons). Green circles show placebo pre-period coefficients; dark red circles show placebo post-period effects; shaded bands are 95 percent confidence intervals.

Figure A.12: Robustness to Not-Yet-Treated Controls: Effect of Medicaid Expansion on Matched Residency Positions per 100,000 Population



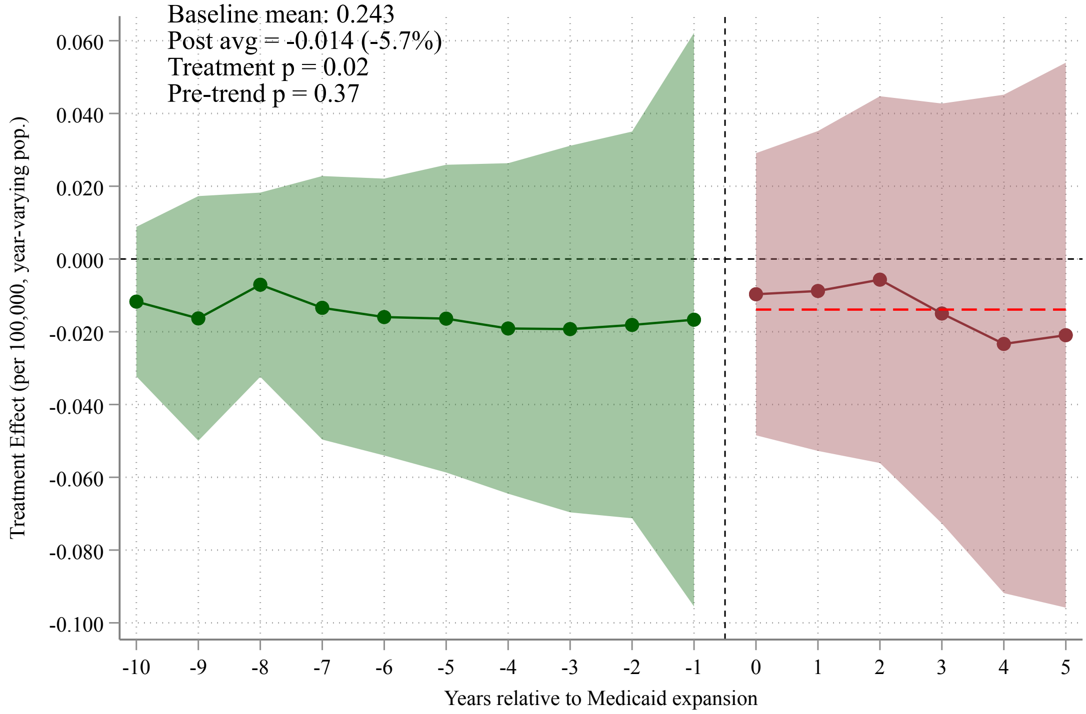
Notes: The figure re-estimates the main event study after dropping never-expansion states, so identification rests solely on variation in the timing of expansion among eventually-treated states (the not-yet-treated design). The outcome is matched positions per 100,000 contemporary (year-varying) population. Because the last-treated cohort has no valid comparison at horizon +5 under the not-yet-treated design, the event study is estimated over horizons 0 to +4. The average post-expansion effect is −0.043 (a 16.6 percent decline relative to the pre-expansion mean of 0.26; s.e. 0.020)—larger than the full-sample effect—but the ten-period window reveals what a five-period window could not: the design's pre-expansion coefficients fail the pre-trends test decisively ($p < 0.001$), so we no longer treat the timing-only estimate as the cleaner alternative and rely on the never-expansion comparison in the main text. Earlier drafts reported this design favorably on a five-period window; the correction is an example of what the longer panel buys. Green circles (years −10 to −1) show pre-expansion coefficients; dark red circles (years 0 to +4) show post-expansion effects; the dashed red line marks the average post-expansion effect. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

Figure A.13: Average Post-Expansion Effects and False-Discovery-Rate Significance Across the Primary Family



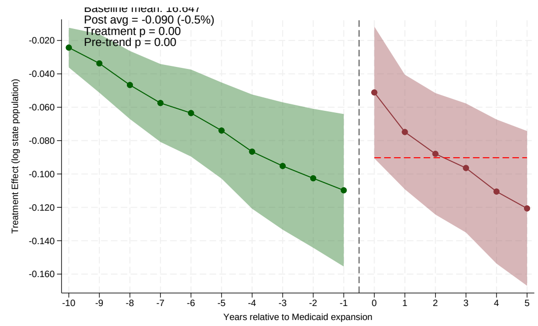
Notes: The figure plots the average post-expansion effect on matched positions per 100,000 contemporary (year-varying) population for each member of the primary family of hypotheses, with 95 percent confidence intervals (point estimate ± 1.96 clustered standard errors). The label beside each marker reports the Benjamini–Hochberg false-discovery-rate q-value under both inference standards: q_{cl} (clustered) and q_{RI} (randomization inference). Under clustered inference no non-mechanism member of the nine-hypothesis family survives a 5 percent false-discovery-rate correction (headline $q = 0.09$); the mechanism rows that do survive are disqualified by failing pre-trends (Section 6.3). Standard errors are clustered at the state level.

Figure A.14: Balanced-Cohort Event Study: 2014 Adopters Only

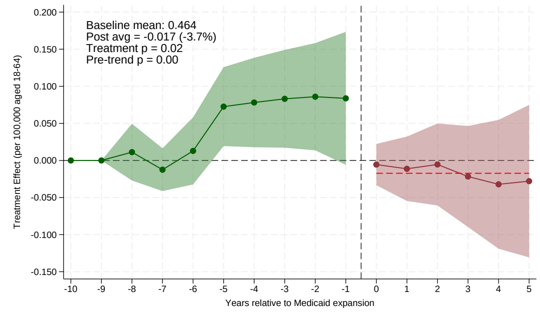


Notes: The figure re-estimates the main event study restricting treatment to the 2014 expansion cohort (the largest wave), with never-expansion states as the comparison group, so that every event-time coefficient is identified from the same fixed set of treated states. The outcome is matched positions per 100,000 contemporary (year-varying) population. The average post-expansion effect is -0.014 (a 5.7 percent decline relative to the pre-expansion mean of 0.24; s.e. 0.028; joint $p = 0.02$), following ten flat pre-expansion coefficients (pre-trend $p = 0.37$)—the single-cohort design reproduces the headline almost exactly. Because cohort composition is held fixed, the widening path here reflects genuine dynamics rather than the changing composition of cohorts across horizons. Green circles (years -10 to -1) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion effects; the dashed red line marks the average post-expansion effect. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

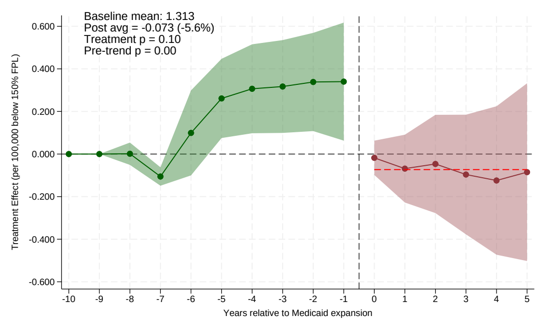
Figure A.15: Denominator Diagnostics: Alternative Deflators and the Population Outcome



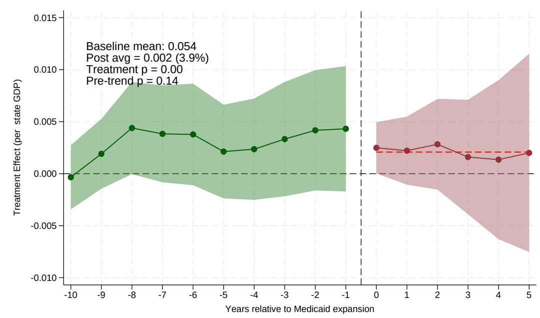
(a) Log contemporary state population as outcome



(b) Matched per 100,000 aged 18–64



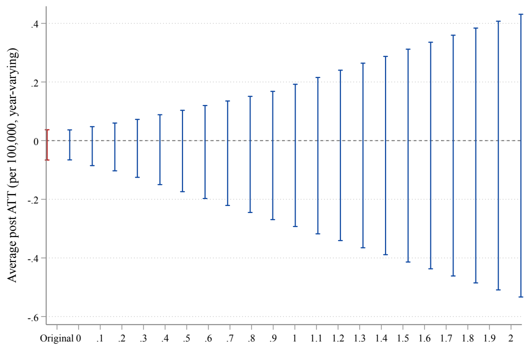
(c) Matched per 100,000 below 150% FPL



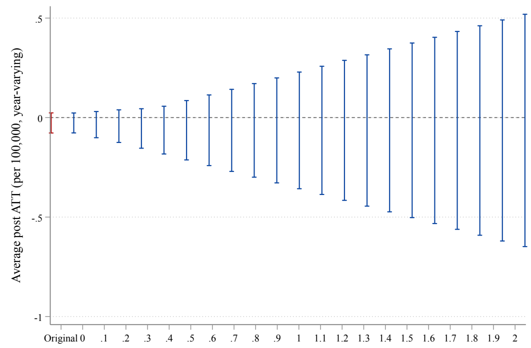
(d) Matched per $billion state GDP

Notes: The figure reports the denominator diagnostics discussed in Section 7.5. Panel (a) uses the log of contemporary state population itself as the event-study outcome: the denominator of the headline ratio trends differentially around expansion, in the direction that biases against finding a decline in per-capita training. Panels (b)–(d) re-estimate the headline event study with matched positions deflated by three alternative scales: the state population aged 18–64 and the population below 150 percent of the federal poverty line (both from the American Community Survey, available 2010–2019), and current-dollar state GDP from the Bureau of Economic Analysis regional accounts (SAGDP2, available for the full 2000–2019 panel)—the non-demographic scale that is not mechanically post-treatment through coverage-induced migration or income effects. Each panel's annotation box reports the average post-expansion effect, its percent-of-baseline equivalent, the joint post-period p-value, and the ten-period pre-trends p-value. The two demographic deflators fail their pre-trends tests; the GDP deflator passes and, together with the contemporary-population headline, brackets zero. Estimates use the imputation estimator of Borusyak, Jaravel, and Spiess (2024) with hospital and year fixed effects, 2010-population weights, and state-clustered standard errors. Green circles show pre-expansion coefficients; dark red circles show post-expansion effects; shaded bands are 95 percent confidence intervals.

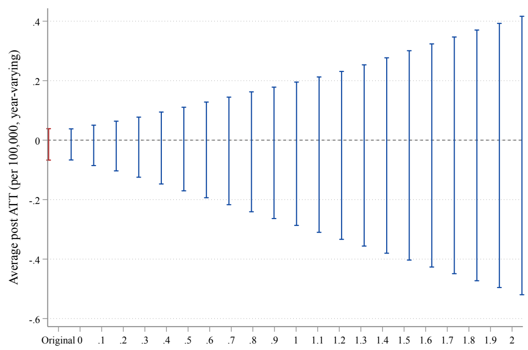
Figure A.16: HonestDiD Sensitivity to Violations of Parallel Trends



(a) Headline (per 100,000, year-varying)



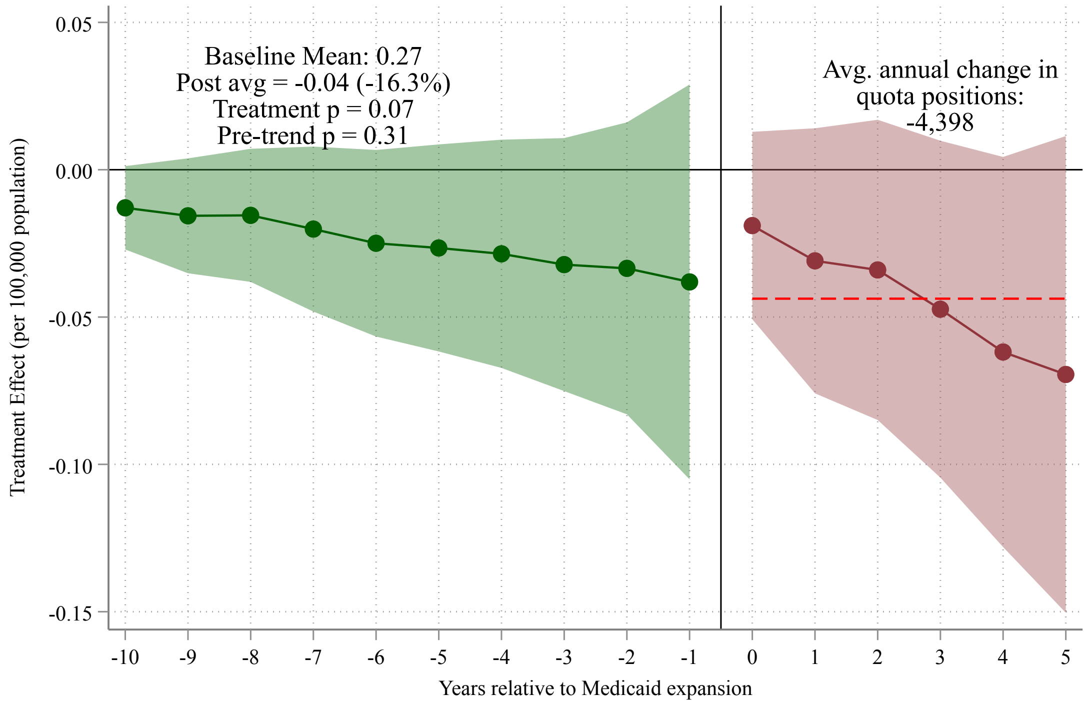
(b) Non-Responsive States



(c) Urban Hospitals

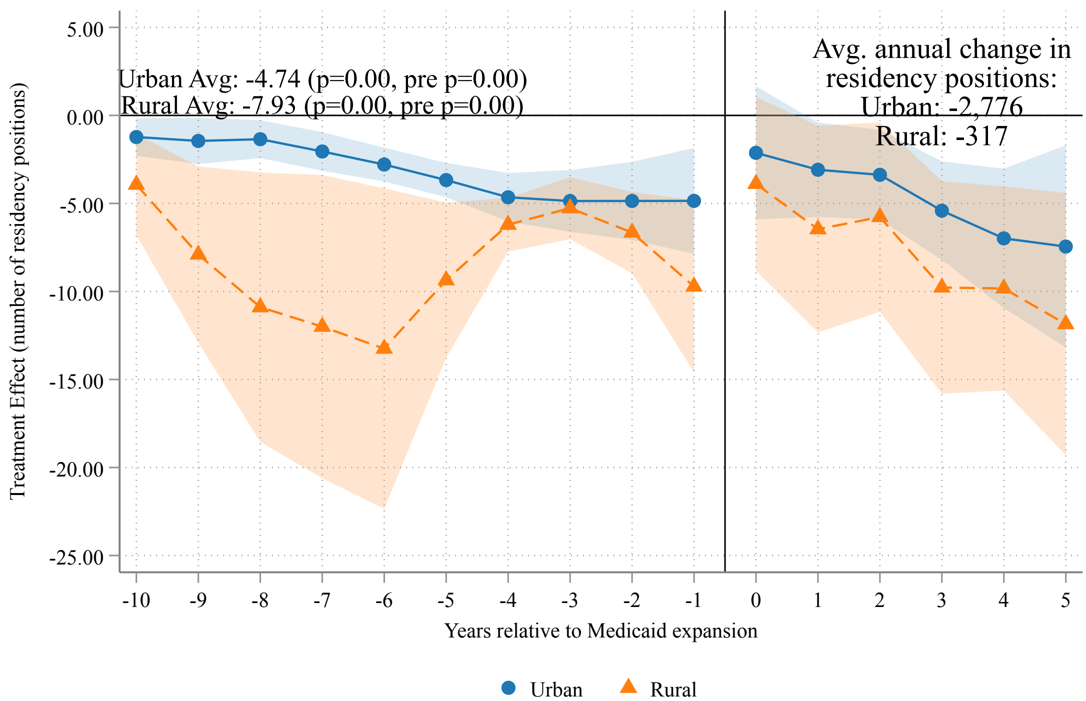
Notes: The figure reports Rambachan and Roth (2023) sensitivity analysis for the average post-expansion effect on matched positions per 100,000 contemporary population, under the relative-magnitudes restriction $\Delta^{RM}(\bar{M})$, which bounds the post-period differential trend by $\bar{M}$ times the largest pre-period differential trend. Each panel plots the robust 95 percent confidence interval for the average post ATT as $\bar{M}$ increases. For the headline effect, the non-responsive mechanism panel, and the urban subsample, the robust confidence interval includes zero even at $\bar{M} = 0$ (exact parallel trends), confirming that the average magnitude is not distinguishable from zero at conventional levels; what the design supports is the sign and the growing dynamic path of the effect, together with the clustered joint tests. The leftmost interval in each panel is the original CI. Estimates use the Borusyak, Jaravel, and Spiess (2024) event-study coefficients with state-clustered standard errors.

Figure A.17: Effect of Medicaid Expansion on Offered Residency Positions (Quota): Event Study Estimates



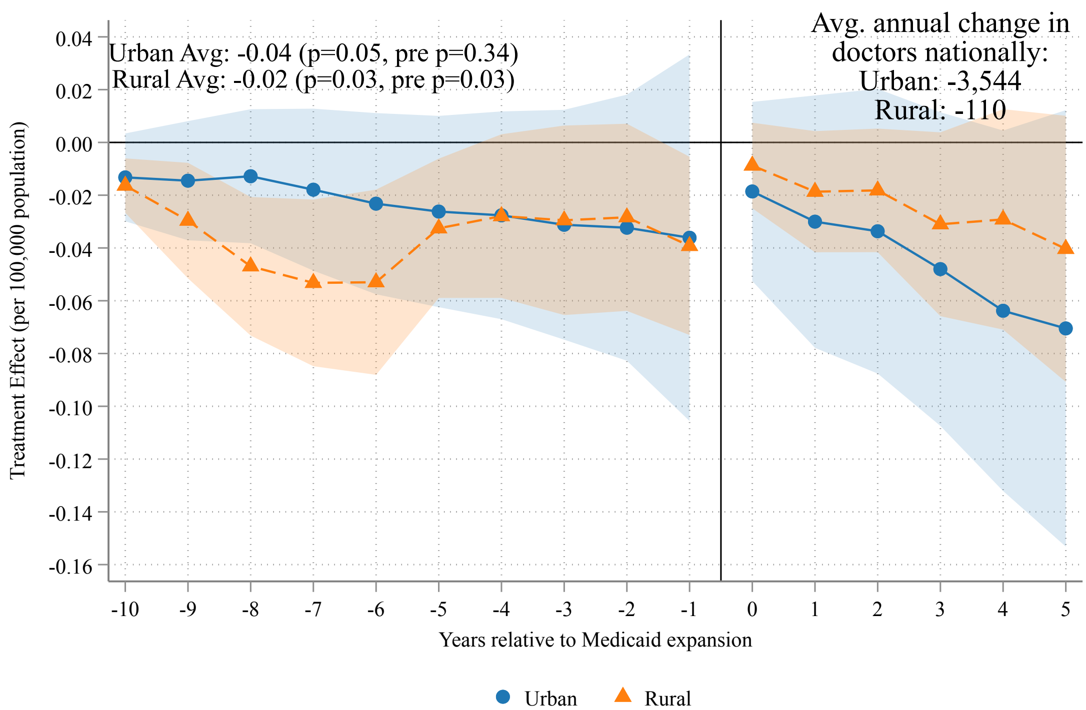
Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), where the outcome is quota positions per 100,000 state population (fixed 2010 population; the year-varying quota estimate of -0.018 is reported in the text). Quota positions are the number of positions offered by residency programs in the NRMP Main Residency Match, prior to the matching algorithm; matched positions (the primary outcome in Figure 5) are the subset of offered positions that are ultimately filled. The annotation box reports the baseline mean, average post-expansion effect, and joint and ten-period pre-trend p-values for this specification; the year-varying quota estimate in the text (-3.0 percent, $p = 0.21$) is the primary offered-positions result and is statistically indistinguishable from zero. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.18: Effect of Medicaid Expansion on Matched Residency Positions by Hospital Location, in Levels: Event Study Estimates



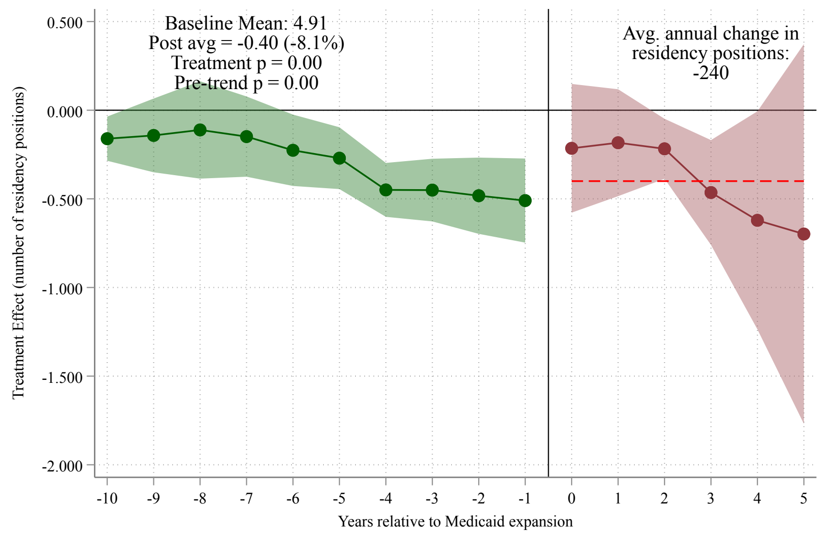
Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), estimated in two split-sample regressions for urban (blue circles) and rural (orange triangles) hospitals; each subsample keeps the never-expansion hospitals of its own type as the comparison group, so each series carries its own pre-expansion path and pre-trends test. The outcome is the absolute number of matched residency positions per hospital (not normalized by population). The average post-expansion treatment effect is -3.62 positions per hospital for urban hospitals (joint $p < 0.001$; pre-trend $p < 0.001$) and -8.44 positions per hospital for rural hospitals (joint $p < 0.001$; pre-trend $p = 0.02$). As with the aggregate levels outcome (Figure A.1), both subsamples fail the pre-trends test in levels, so these magnitudes corroborate the direction of the effect only. Translating to aggregate counts, expansion states experienced average annual reductions of approximately 2,120 matched positions at urban hospitals and 338 at rural hospitals. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

Figure A.19: Effect of Medicaid Expansion on Offered Residency Positions (Quota) by Hospital Location: Event Study Estimates

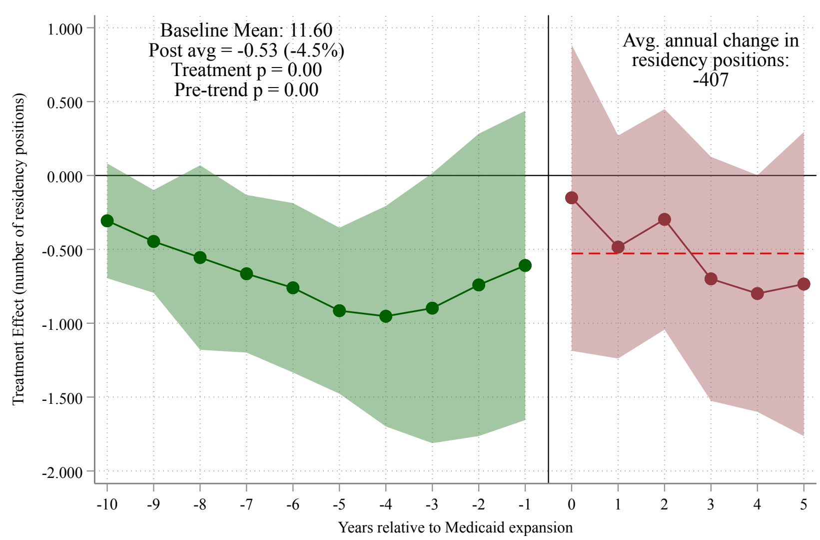


Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), estimated in two split-sample regressions for urban (blue circles) and rural (orange triangles) hospitals; each subsample keeps the never-expansion hospitals of its own type as the comparison group, so each series carries its own pre-expansion path and pre-trends test. The outcome is quota positions per 100,000 state population (fixed 2010 population), where quota measures positions offered by programs prior to the matching algorithm. The average post-expansion treatment effect is -0.031 for urban hospitals (joint $p = 0.03$; pre-trend $p = 0.10$) and -0.026 for rural hospitals (joint $p = 0.03$), similar in magnitude across the two groups. The rural subsample fails its own pre-trends test ($p < 0.001$), so the rural series should be read descriptively. These estimates correspond to average annual reductions of approximately 2,452 offered positions at urban hospitals and 118 at rural hospitals. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

Figure A.20: Effect of Medicaid Expansion on Matched Residency Positions by Specialty Type, in Levels: Event Study Estimates



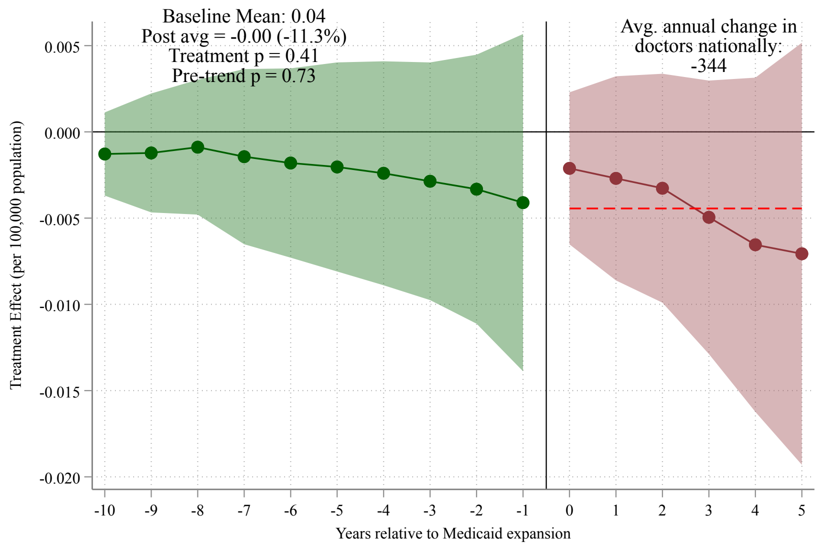
(a) Non-Primary Care Positions



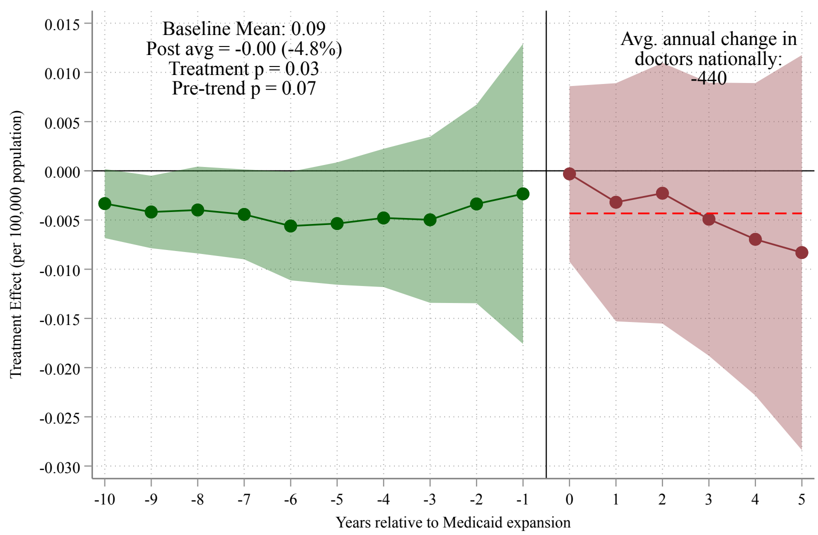
(b) Primary Care Positions

Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), estimated separately for non-primary care (top panel) and primary care specialties (bottom panel). The outcome is the absolute number of matched residency positions per hospital (not normalized by population). Primary care includes family medicine, internal medicine, and pediatrics. The average post-expansion treatment effect is −0.37 positions per hospital for non-primary care (−8.6 percent relative to the pre-expansion mean of 4.37; $p < 0.01$), corresponding to an average annual reduction of approximately 155 matched positions nationally. For primary care, the average treatment effect is −0.95 positions per hospital (−10.0 percent relative to the pre-expansion mean of 9.52; $p < 0.01$), corresponding to an average annual reduction of approximately 562 positions. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.21: Effect of Medicaid Expansion on Offered Residency Positions (Quota) by Specialty Type: Event Study Estimates



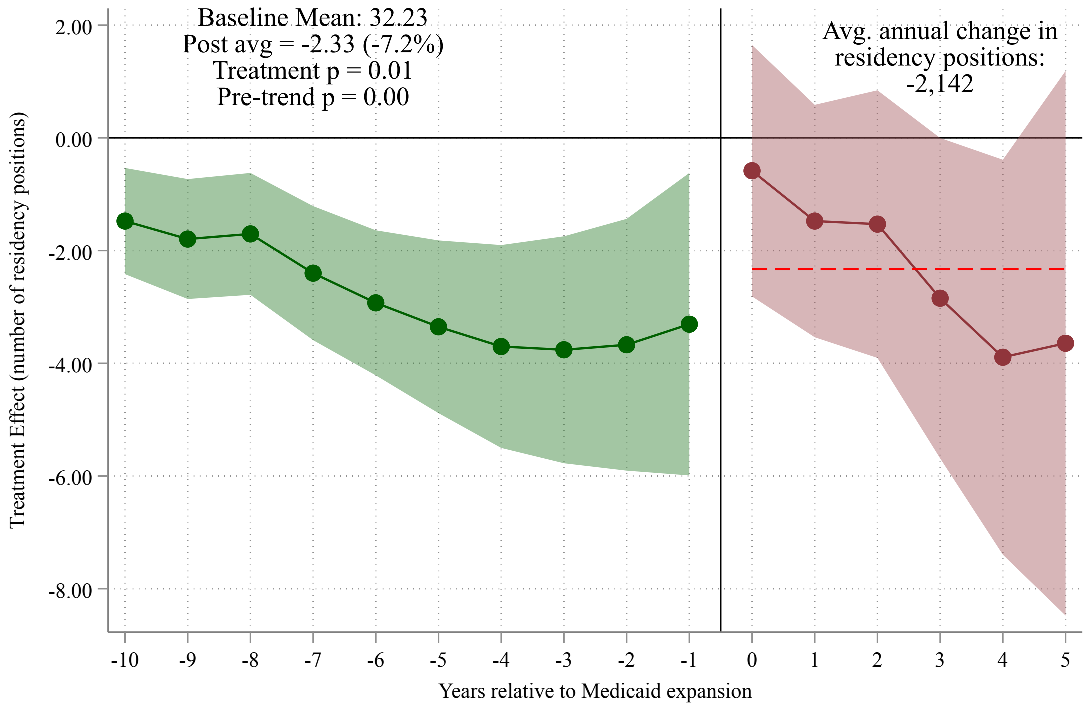
(a) Non-Primary Care Positions



(b) Primary Care Positions

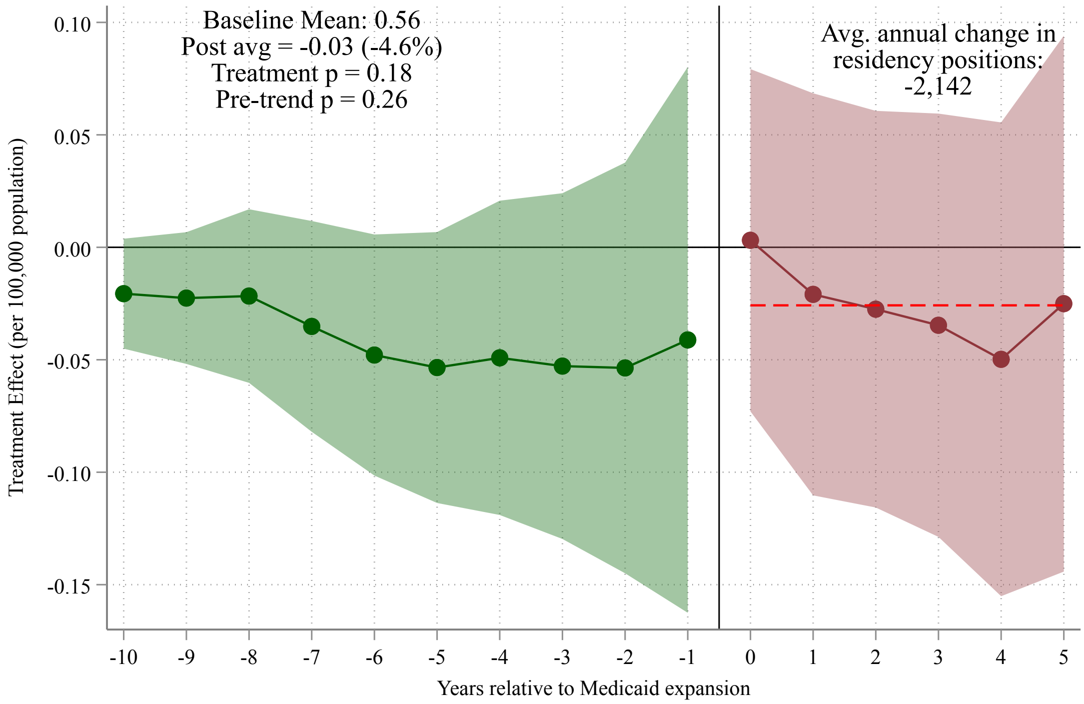
Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), estimated separately for non-primary care (top panel) and primary care specialties (bottom panel). The outcome is quota positions per 100,000 state population, where quota measures positions offered by programs prior to the matching algorithm. For non-primary care, the average post-expansion treatment effect is -0.00 (-9.7 percent relative to the pre-expansion mean of 0.04; $p = 0.12$), corresponding to an average annual reduction of approximately 193 offered positions nationally. For primary care, the average treatment effect is -0.01 (-9.3 percent relative to the pre-expansion mean of 0.07; $p < 0.01$), corresponding to an average annual reduction of approximately 554 offered positions. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.22: Effect of Medicaid Expansion on Matched Residency Positions (Unweighted): Event Study Estimates



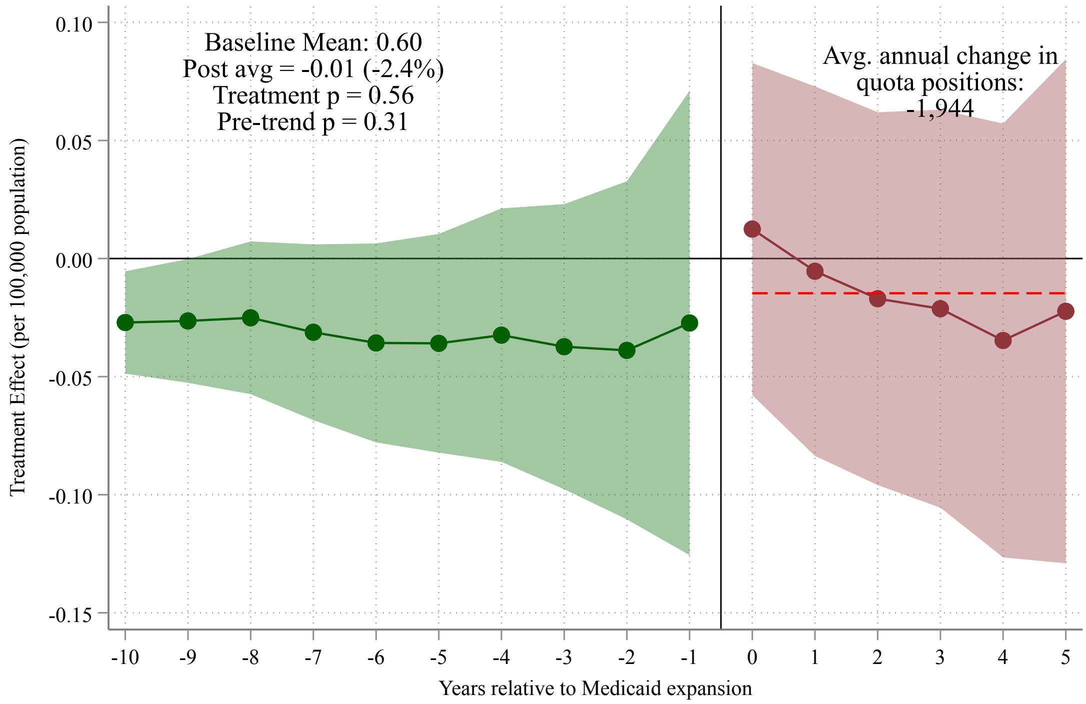
Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), where the outcome is the number of matched residency positions. This specification is estimated without population weights and includes 2010 state population as a control. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. Green circles (years -10 to -1) show pre-expansion coefficients. Dark red circles (years 0 to $+5$) show post-expansion treatment effects. The annotation box reports the baseline mean, average post-expansion effect, and joint and ten-period pre-trend p-values; the levels outcome fails parallel trends on the full panel, so this figure corroborates direction only (Section 7.5), and we no longer translate levels estimates into national position counts. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.23: Effect of Medicaid Expansion on Matched Residency Positions per 100,000 Population (Unweighted): Event Study Estimates



Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), where the outcome is matched residency positions per 100,000 state population. This specification is estimated without population weights and includes 2010 state population as a control. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. Green circles (years -10 to -1) show pre-expansion coefficients. Dark red circles (years 0 to $+5$) show post-expansion treatment effects. The annotation box reports the baseline mean, average post-expansion effect, and joint and ten-period pre-trend p-values; the estimate is statistically indistinguishable from zero, consistent with the unweighted cell of the specification grid discussed in Section 7.2. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.24: Effect of Medicaid Expansion on Residency Quota Positions per 100,000 Population (Unweighted): Event Study Estimates



Notes: The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), where the outcome is residency quota (offered) positions per 100,000 state population. This specification is estimated without population weights and includes 2010 state population as a control. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. Green circles (years −10 to −1) show pre-expansion coefficients. Dark red circles (years 0 to +5) show post-expansion treatment effects. The annotation box reports the baseline mean, average post-expansion effect, and joint and ten-period pre-trend p-values; the estimate is statistically indistinguishable from zero. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

C The Medicaid GME Formula Classification

The mechanism analysis in Section 6.3 classifies each state by whether its Medicaid program makes graduate medical education payments through a formula that scales with Medicaid patient volume, so that a coverage expansion mechanically raises GME revenue. We code the payment rules in force during the 2014–2019 treatment window from the AAMC fifty-state survey of Henderson (2016), and we release the state-by-state coding file, with source notes, in the replication package. Five states required judgment calls (Maryland, Minnesota, Montana, Iowa, and New Mexico); Section 6.3 shows the cross-arm difference is robust to flipping each of them.

California carries the largest weight of any state in the estimation, and its coding as non-paying deserves an explicit defense. Medi-Cal made no fee-for-service GME payments during the study window, which is the basis of the survey coding. California's state GME support instead ran through appropriated grant programs: Song-Brown, funded from the state general fund and small before a one-time $100 million augmentation beginning in fiscal year 2017–18, and CalMedForce, funded at roughly $40 million per year from Proposition 56 tobacco-tax revenue with first grants in 2018–19 (California Health Care Foundation 2024; Legislative Analyst's Office 2019). Neither program's funding moves with Medicaid enrollment or utilization; both are fixed appropriations allocated by grant competition. Nor did managed care embed a volume-scaled GME formula: capitation rates paid to Medi-Cal plans carry no identifiable GME component tied to the training hospital, and supplemental support for the designated public hospitals flowed through waiver-financed pools budgeted against uncompensated care rather than resident counts or Medicaid discharge volume. Timing reinforces the coding: the two channels that did expand state GME dollars begin in 2017–2019, late in the post window, and are not Medicaid-financed, so if anything they push against finding a contraction in California and make the non-paying coding conservative for the mechanism test.

The residual concern is that the waiver pools functioned as de facto volume-responsive GME support, in which case California would be misclassified. Because California carries the largest weight in the estimation, this possibility is one more reason Section 6.3 reads the cross-arm contrast descriptively and rests no mechanism claim on the classification.

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